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A Base Integer Programming Model and Benchmark Suite for Liner-Shipping Network Design

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The liner-shipping network design problem is to create a set of nonsimple cyclic sailing routes for a designated fleet of container vessels that jointly transports multiple commodities. The objective is to maximize the revenue of cargo transport while minimizing the costs of operation. The potential for making cost-effective and energy-efficient liner-shipping networks using operations research (OR) is huge and neglected. The implementation of logistic planning tools based upon OR has enhanced performance of airlines, railways, and general transportation companies, but within the field of liner shipping, applications of OR are scarce. We believe that access to domain knowledge and data is a barrier for researchers to approach the important liner-shipping network design problem. The purpose of the benchmark suite and the paper at hand is to provide easy access to the domain and the data sources of liner shipping for OR researchers in general. We describe and analyze the liner-shipping domain applied to network design and present a rich integer programming model based on services that constitute the fixed schedule of a liner shipping company. We prove the liner-shipping network design problem to be strongly NP-hard. A benchmark suite of data instances to reflect the business structure of a global liner shipping network is presented. The design of the benchmark suite is discussed in relation to industry standards, business rules, and mathematical programming. The data are based on real-life data from the largest global liner-shipping company, Maersk Line, and supplemented by data from several industry and public stakeholders. Computational results yielding the first best known solutions for six of the seven benchmark instances is provided using a heuristic combining tabu search and heuristic column generation.

Keywords: liner shipping; mathematical programming; network design

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1. A Benchmark Suite for Research on Liner-Shipping Network Design Problems

Operations research (OR) is widely used within the transportation sector to provide a cost-efficient and competitive organization. However, application of OR within containerized liner shipping is scarce (Christiansen, Fagerholt, and Ronen 2004). The potential impact of OR on this billion dollar industry is enormous, especially given the large concentration of players in the business. Maritime shipping produces an estimated 2.7% of the world's CO₂ emissions, of which 25% is attributable to container ships

alone (The World Shipping Council 2009). An energy-efficient liner-shipping network is becoming increasingly important to all stakeholders. OR can help in designing effective and energy-efficient liner-shipping networks to mitigate the carbon footprint of the liner-shipping industry. We believe that the lack of OR within liner shipping is partly because of barriers for new researchers to engage in the liner-shipping research community. Constructing mathematical models and creating data for computational results requires profound knowledge of the domain and data sources. The benchmark suite presented in this paper aims to make liner-shipping network design problems (hereafter LSNDP) approachable for

the research community in general. We wish to create a platform where methods can be compared on a set of known data instances. By disseminating our knowledge of the liner-shipping domain into real-world network data instances for mathematical programming, we hope to diminish this formidable hurdle. Our hope is to enable research on the field of liner-shipping network design to develop like the vehicle-routing problems (VRP) have through the benchmark instances of Solomon (1987). The benchmark suite is seen as the root of a tree where new branches will appear as our ability to solve more complex interpretations of the liner-shipping problem grows.

The initial model of the diverse field of VRP models was the capacitated vehicle-routing problem (CVRP). We present a reference model for the LSNDP, which is an extended version of the model in Alvarez (2009). The reference model may be used for comparing various methods applied to the problem and serve as an inspiration to future model development. Our goal is to enable the benchmark suite to support future model and method development, and hence the data of the benchmark suite encompass several attributes, such as transit times not applicable to the reference model. In §2, relevant literature within the field of liner-shipping economics will underpin the importance of various costs and restrictions within network design. A review of the literature within OR and the various models presented is provided. We review methods for solving liner-shipping network design related problems and the computational results in the literature to illustrate the computational hardness of the problem. Section 4 provides an introduction to the liner-shipping network design domain. The domain discussion is complemented by the strategic business and domain knowledge of one of the major global operators within liner shipping. We discuss the data provided in the benchmark suite in relation to the data needed in current mathematical models and the open research questions we try to support. In §5, we discuss the data objects and data generation. The goal for the instances generated is to capture real-life cost structures, trade imbalances, market shares, and scale for a global liner-shipping company. We discuss the diversity of instances needed to qualify and challenge development of both exact and heuristic methods for solving the LSNDP. Finally, we solve the generated instances by using a new tabu search heuristic based on the framework by Alvarez (2009). The algorithm has been extended to handle more complex routes (so-called butterfly routes) and to ensure a weekly or biweekly service frequency. Moreover, an improved MIP neighborhood is used to generate new candidate routes. Computational results are presented in §6 to qualify the benchmark suite

with experimental results and establish the currently best-known solutions for the liner-shipping network design problem and hopefully spur competition and interest into our field of research. Section 8 contains conclusions and directions for further research into liner-shipping network design problems. Sections 4 and 5 are a shortened version of the technical report (Løfstedt et al. 2011). The reader is referred to this report for additional information.

2. Literature Review

The two papers, Christiansen, Fagerholt, and Ronen (2004) and Christiansen et al. (2007) provide excellent surveys of OR in maritime transportation. They point to the limited research of planning problems in liner shipping in contrast to the size and the possible impact of optimization within the industry. There are a number of contributions on the individual economic and structural factors in designing liner-shipping networks, but no comprehensive description of the general liner-shipping domain related to network design has been published to date. One contribution of the present paper is to give an overview of these factors related to the LSNDP. In the first part, we give an overview of contributions related to some of the factors affecting network design such as *network configuration*, *bunker price*, *transit time*, *competitive position*, *repositioning of empty containers*, *frequency*, *port call sequence*, *schedule*, and *fleet deployment*. In the second part we review contributions on the LSNDP.

The *network configuration* of carriers is explored in Notteboom (2004), concluding that a global network will not have a pure hub-and-spoke structure or a pure multiport structure. The economy of deploying super panamax vessels on either a multiport or hub-and-spoke network structure is investigated in a case study by Imai, Shintani, and Papadimitriou (2009). They conclude that the multiport structure is superior for the Asia–North America and Asia–Europe trades, whereas the hub-and-spoke structure is advantageous in the European trades.

The impact of *bunker price* on the network configuration of liner-shipping companies has been explored by Stopford (2009), Notteboom and Vernimmen (2009), and Cariou (2010). Seen from the carriers' point of view, reducing speed and making effective use of the capacity deployed in the network is economically attractive, because the general analysis of vessel costs by Stopford (2009) reveals bunker fuel as the dominant cost in operating a liner-shipping network. According to Notteboom and Vernimmen (2009), managing the bunker consumption in the network gives the carriers strong incentive to reduce speed, deploy additional vessels to services, and increase buffer time in the schedule to avoid having

to increase speed to accommodate for delays and port congestion. The incentive to reduce speed depends on the actual bunker price, because the additional vessels deployed to maintain the frequency come at an increased capital cost. Cariou (2010) argues that a bunker price exceeding 350 USD per ton will ensure the sustainability of “slow steaming,” which means sailing at a reduced average speed. Cariou (2010) calculates the CO₂ reductions from slow steaming from 2008 to 2010 to 11% attesting that slow steaming is a very effective way of reducing the carbon footprint of the liner shipping industry. Recently, Wang and Meng (2012c) investigate sailing speed optimization for each individual port-to-port voyage on a liner-shipping service to reduce bunker consumption using an outer-approximation method.

In liner shipping there is an inherent trade-off between reducing bunker consumption through speed reduction and achieving competitive *transit time* for cargo. Notteboom (2006) explores the time factor in liner-shipping network designs related to transit time of cargo-routing and schedule reliability. An important analysis on the relation between the number of port calls on a service and competitive transit times is reported. The conclusion is that the time spent at ports is very significant, and hence the number of port calls is decisive for the transit time of direct connections at the end points of a service. Wang and Meng (2011) evaluate the schedule design along with container routing for a fixed network with predefined paths for container shipments to minimize transshipment cost and transit time.

Gelareh, Nickel, and Pisinger (2010) explore the possible *competitive positions* of a carrier in a market with an incumbent carrier. The relation between time and cost on the market share is modeled and investigated. Gelareh, Nickel, and Pisinger (2010) underpin transit times and level of service as important factors in the construction of a liner-shipping network design. Wang and Meng (2012b) propose a tactical model for schedule design capturing the trade-off between speed optimization and transit time levels, taking into account time uncertainty at sea and in the port times. We believe there will be a surge in sailing speed optimization taking level of service and schedule reliability into account where the liner shipping service network has already been fixed to a large degree. We are not aware of published models of the LSNBP incorporating distinct cargo transit times or level of service, but we believe that future research on the LSNBP will incorporate these factors.

Another important factor in a liner-shipping network is *repositioning of empty containers* due to trade imbalances. Shintani et al. (2007) design a model for a single service of a carrier. The experimental results indicate that empty repositioning is significant for the

port-calling sequence and the cargo-handling costs incurred. Dong and Song (2009) investigate the proportion of empty repositioning given current global container trade using the existing global liner shipping network. Their results conclude that 27% of all container traffic is empty repositioning. Brouer, Pisinger, and Spoorendonk (2011) show that joint optimization of demanded cargo and empty repositioning in a fixed network is viable. Meng and Wang (2011) incorporate empty repositioning and port productivity into the evaluation and selection of candidate shipping routes. Empty repositioning is so far not incorporated into the cargo allocation in models where the set of candidate routes are not fixed as in the LSNBP.

Liner shipping generally adheres to a schedule based on *weekly or biweekly frequency of port calls* and deploying vessels of similar characteristics to each service (Stopford 2009; Notteboom 2004, 2006). In Alvarez (2012) the cost of the practice of a weekly frequency is questioned, and level of service expressions are formulated to investigate the impact of maintaining a weekly frequency as opposed to another fixed frequency. Several contributions on the optimal *port-calling sequence* may be found for a single route, e.g., Lu (2002); Chu, Kuo, and Shieh (2003); Shintani et al. (2007); Hsu and Hsieh (2007).

Scheduling of liner-shipping services is a related-optimization problem. An example of scheduling with fixed port call sequence may be found in Yan, Chen, and Lin (2009), as well as the more recent papers of Wang and Meng (2011, 2012b), whereas Agarwal and Ergun (2008b) decides scheduling to a certain weekday and routing simultaneously.

The *fleet deployment problem* (FDP) is closely related to LSNBP because fleet deployment is often implicitly considered in the network cost. FDP assumes that the service in terms of actual vessel voyages is fixed, and hence, that the routing is already decided upon. FDP decides on how to assign the carrier’s own vessels to the actual vessel voyages along with the options of chartering in (leasing) vessels or chartering out (forward leasing) own vessels to minimize the cost of maintaining the schedule. Fleet deployment is generally described in Christiansen et al. (2007), with reference to the main paper within fleet deployment, Powell and Perakis (1997). Fagerholt, Johnsen, and Lindstad (2009) is a more recent paper and case study on liner-shipping fleet deployment using Höegh autoliners as a case study. Another recent paper by Gelareh and Meng (2010) considers a liner-shipping fleet deployment problem, where the frequency of sailing is decided upon in relation to demand coverage and speed on the individual voyage. The paper gives a thorough review of fleet deployment literature to date. The model is solved

using a MIP solver for three and four services using three to five vessel types. Recently, the fleet deployment problem in conjunction with transit time levels was described by Wang and Meng (2012a). Meng and Wang (2012) expanded Wang and Meng (2012a) with weekly frequency considerations.

Literature on models and methods for *liner shipping network design* within OR is still limited, although recent years show increasing activity in the field. Kjeldsen (2011) provides a classification scheme for routing and scheduling within liner shipping. The classification scheme entails 24 references in total for routing and scheduling problems within liner shipping. The classification scheme of Kjeldsen (2011) shows that there is a large variation in the models with respect to cost structures, constraints, and scope unveiling that we are dealing with a young research field with many possibilities of scoping network design to various decisions such as scheduling, fleet sizing, fleet deployment, route generation, speed optimization, and many more.

An early contribution is that of Rana and Vickson (1991), who present a model for liner shipping with nonsimple routes by using a head and backhaul structure. The model does not allow transshipments, which are at the core of liner shipping today. Fagerholt (2004) develops a model and solution method for a regional carrier along the Norwegian coast. The model assumes the carrier loads at a single port and finds optimal routes of vessels to service the unloading facilities. The problem may be dealt with as a VRP, given that a designated depot is known and transshipments are not allowed. The solution method is a MIP solver. Similarly, Karlaftis, Kepaptsoglou, and Sambracos (2009) solve a problem for the region of the Aegean Sea using a genetic algorithm related to the coastal freight problem described in Sambracos et al. (2004). These models do not deal with the important concept of transshipments at multiple ports and the resulting interaction between different services.

Agarwal and Ergun (2008b) were the first to include a *weekly frequency constraint* by grouping vessels into vessel classes in the *simultaneous ship-scheduling and cargo-routing model*. The model creates routings for a set of vessel classes with a rough schedule. The weekly frequency constraint determines the number of vessels deployed to a service according to the service duration, and they introduce a time-space graph spanning each weekday to reflect the availability of transportation on a certain weekday. The model allows for transshipments, but the cost of transshipments cannot be derived, and hence the overall cost structure excludes the important cost of transshipment. The time-space graph gives a temporal aspect in terms of a rough schedule based on weekdays, but the time-space graph is not utilized

to reflect transit time restrictions. Blander Reinhardt and Pisinger (2012) present a model that is among the first to include transshipment cost and present a branch-and-cut method for solving a liner-shipping network design problem with butterfly routes to optimality. The model selects a route for each individual vessel in a fleet. This results in a distinct network layer for each vessel, and the grouping of vessels to a single service is not explicit in the model. The configuration is suitable for smaller carriers. Global carriers tend to group a set of vessels with similar characteristics to a single service to reduce the complexity of the network design (Stopford 2009; Notteboom and Vernimmen 2009) and to provide simpler schedules to their customers.

Alvarez (2009) is the most recent publication on LSNDP considering *the joint routing and fleet deployment model*. The model bundles a service with a vessel class, the number of vessels deployed to the service, a target speed, and a nonsimple cyclic port sequence. The cost of a service in the model reflects the deployment cost of the vessels in the service and the estimated bunker consumption, adjusting for the difference in bunker consumption when sailing at different speeds or idling at a port. The total fleet cost accounts for whether vessels are deployed to services or chartered out at market prices. Cargo revenues and handling costs are accounted for in the model, along with a penalty for cargo that has been forfeited because of insufficient capacity or insufficient revenue in the view of total network cost. The model assumes a planning horizon with stable demands and does not impose restrictions on the frequency of service or the actual scheduling of the services. A case study with discussion on data generation and port/vessel class incompatibilities is provided. The transshipment cost of nonsimple routes is, however, not calculated coherently, because the model cannot detect such a transshipment.

Agarwal and Ergun (2008b) prove the NP-completeness of *the simultaneous scheduling and cargo-routing problem*. The LSNDP, in general, may be viewed as a VRP with split pickups, split deliveries, multiple cross docking, no depots, and a heterogeneous vessel fleet, and hence the LSNDP is strongly NP-hard, as proven in §6.4. The methods deployed to solve the problem are mainly heuristics based on integer programming and decomposition techniques. The branch-and-cut method of Blander Reinhardt and Pisinger (2012) and a MIP formulation of Alvarez (2009) solve smaller instances to optimality. Remaining solution methods are heuristic methods. An overview of the reviewed methods on liner-shipping network design is given in Table 1.

Computational results are scarce and based on individual data sets, but it is clear from the literature that solving large-scale instances is a hard task

Table 1 Overview of the Models on Liner-Shipping Network Design Published in International Journals

Article	Routes	Fleet	Routing	Method	Optimal	Constraints	Vessels/ports
Rana and Vickson (1991)	Multiple	Heterogen	In/out bound	Lagrange, Benders	No	Capacity, time, connected	3 v, 20 p
Shintani et al. (2007)	Single	Vessel class	In/out bound	Genetic algorithm	No	Empty repositioning, bunker cost, and handling cost	3 v, 20 p
Karlaftis, Kepaptsoglou, and Sambracos (2009)	Multiple	Heterogen	Simple cycle–VRP	Genetic algorithm	No	Capacity and time windows on demands	Not fixed, 26 p
Fagerholt (2004)	Multiple	Heterogen	Simple cycle–VRP	Route generation/MIP solver	Yes	Capacity, route duration	20 v, 40 p
Blander Reinhardt and Pisinger (2012)	Multiple	Heterogen	Butterfly cycle	Branch and cut	Yes	Capacity, time, connected, transshipment	6 v, 15 p
Agarwal and Ergun (2008b)	Multiple	Vessel class	Simple cycle	Greedy, column generation, Benders	No	Weekly frequency, transshipment	50 v, 10 p
Alvarez (2009)	Multiple	Vessel class	Nonsimple	MIP solver/tabu search	Yes/No	Transshipment, bunker	100 v, 7 p (MIP solver), 120 p (Tabu search)

Notes. Routes: refers to whether the problem is defined for a single or multiple routes/services. A Fleet is homogeneous if only a single type of vessel (in terms of capacity and technical specifications) and heterogeneous for multiple vessel types. Routing denotes the way routes are perceived and hence constructed by the model, see §4.2.3. Method refers to the algorithmic solution method, Optimal to whether the method proves optimality of a solution, Constraints are a summary of the constraints included. Lastly, Vessels/ports states the number of vessels and ports in the computational study.

even for heuristics. The benchmark suite presented in this paper will aid comparison between different methods if the public benchmark data is used for computational results. In Christiansen and Fagerholt (2011), the importance of creating public benchmark instances for maritime transportation problems is advocated to drive basic research on maritime optimization problems forward.

Alvarez (2009) identifies several problems with exact optimization methods. Firstly, the feasible solution space of a single rotation is vast. Combining this with a vector of different capacities and cost for every feasible rotation increases the feasible solution space significantly. Because of economies of scale, a solution to the LP-relaxation will prefer to use a fraction of large vessels, leading to highly fractional solutions. This is amplified by the cost structure, where the majority of the network cost is placed on the integer variables, whereas the revenue follows the fractional variables representing the cargo flow. Therefore, convergence of decomposition techniques such as column generation might be slow.

Recent developments in modelling the LSNDP have tried to overcome this division by coupling service generation with the cargo load, as seen in Jepsen et al. (2011) and Kjeldsen (2012). No computational results are reported for Jepsen et al. (2011) because the pricing problem has a very complex structure that cannot be solved efficiently using current state-of-the-art methods such as the resource-constrained shortest-path problem. Kjeldsen (2012) solves the pricing problem heuristically and some limited computational results are reported that confirm the hardness of this problem even for heuristic methods.

3. Contribution

The present paper gives an overview of the domain of liner-shipping network design related to mathematical formulations of the LSNDP and other network-related optimization problems. It is evident that modelling the LSNDP in a compact formulation that can exploit state-of-the-art mathematical programming techniques is still a challenge for our community, and we expect further development in this area. In this paper we present a reference model, which is an extension of Alvarez (2009), where transshipment on butterfly routes is correctly accounted for in the objective function without introducing increased complexity and (bi)weekly frequency is imposed in the route generation. The model presented is a simplified but realistic description of the LSNDP as described in this paper. We believe that the model, along with the benchmark suite, can create a platform for the development of heuristic and exact solution methods. We provide a heuristic solution method for solving the LSNDP and the first computational results for our benchmark suite. Any model adhering to the constraints and the objective of the reference model of this paper should be able to compare itself to the computational results presented and improve the research direction of the LSNDP. The benchmark suite can cater for extensions such as service-level requirements, port productivity, and sailing speed optimization to support future research directions within the field of mathematical models of the LSNDP. The benchmark suite is also applicable to other related and promising research problems within liner-shipping network design, where the routes or the cargo flows have been fixed to some

extent as seen in, e.g., Meng and Wang (2011); Wang and Meng (2011).

4. Domain of the Liner-Shipping Network Design Problem

The market and business of liner shipping is thoroughly described and analyzed in Stopford (2009) and Alderton (2005). Complemented by the experience of network planners and optimization managers at Maersk Line, this section introduces the business context of liner-shipping network design.

4.1. The Liner-Shipping Business

The business of liner shipping is often compared to public transit systems such as bus lines, subways, and trains. A *service* is a *round trip* sailed at a given *frequency*. The schedule can be consulted to find the next call of the service at a port. The *round trip* is the set of designated stops, which for a liner-shipping company is a set of *ports*. The round trip in liner shipping may be divided into a *headhaul* and a *backhaul* to distinguish between different demands according to their regional specifications. The headhaul is usually the demand-intensive direction. Some services are dedicated to connecting many origins and destinations, whereas others serve a smaller distinct market. In liner shipping a distinction is made between *trunk services* serving central main ports with several demands and *feeder services* serving a distinct market typically visiting a single main port and a set of smaller ports. As in public transit, a transport may include the use of several services to connect between the origin and the destination of the transport. In liner shipping we refer to transits as *transshipments*. A *fleet of vessels* with varying capacity and speed is deployed to the services according to the demand. The *transit time* of cargo denotes the time cargo travels from origin to destination. Transit time is counted in days, and transit times may vary from a single day to 90 days.

4.2. Network

The network must be *competitive* and *efficient*. A *competitive* network may accommodate several routings for one origin-destination pair varying on transit time and cost. Most global liners provide several itineraries for an origin-destination pair by end-to-end services, as well as transshipment services with different transit times and freight rates. For a network to be *competitive* it must offer low transit times and few transshipments. A *competitive* network serves the main ports of a region frequently with good connections to feeder ports with a high schedule reliability. An *efficient* network facilitates transshipments at terminals with high crane productivity and container capacity to minimize the cargo-handling time. Transshipments are also used to get effective utilization of

vessel capacity such that the trunk services are fed by several feeder services and direct cargo along the trunk line. Finally, empty containers must be effectively repositioned to ensure availability of containers at the origins of the cargo. This is especially critical for reefer containers because several regions such as South America and New Zealand have a large export of refrigerated goods, but an insignificant import of refrigerated goods. Examples of different liner shipping networks are seen in Notteboom (2004), which are included in Figure 1. Notteboom (2004) argues that global liners are multilayered networks of different types because they are all competitive for particular circumstances.

4.2.1. Value Propositions. A specific liner-shipping company is referred to as a *carrier*, whereas the owner of a certain cargo is denoted a *shipper*. The competitive position of a carrier is a combination of port coverage, price, transit time, transshipments, schedule reliability, and, in recent years, corporate social responsibility of the company, where environmentally friendly transport has received a lot of attention. The *freight rate* of transporting some cargo depends not only on the actual network cost, but also on the transit time, the container type needed (e.g., a refrigerated container), special regulations (restricted and dangerous goods), the number of transshipments, cabotage regulations, and, naturally, the relation between supply and demand for cargo transport on the connection in question. In addition, the price covers the administrative overhead incurred by the carrier, comprising approximately 30% of the total cost (Stopford 2009). Notteboom (2006), Notteboom and Vernimmen (2009), and Stopford (2009) discuss the importance of transit time in pricing cargo transportation as the inventory-holding cost is paid by the shipper until the cargo is delivered (Notteboom 2006). Furthermore, some goods are perishable and the time to market becomes crucial. It is becoming increasingly important for companies to ensure transport with the lowest CO₂ emission, and hence slow steaming (in particular on backhaul connections) to reduce network bunker consumption and overall CO₂ emission.

4.2.2. Forecast and Planning Horizon. A demand forecast for a given planning horizon is crucial to network design because the ideal network has a perfect fit between demand and capacity (Stopford 2009). The demand for container transport fluctuates over a year with seasonal variance and peaks at certain times of year. These peaks may be *regional*, if they are related to a crop (e.g., bananas or lemons), or *global* (e.g., Christmas). Hence, the network design is rarely stable over a yearly period. Some structure is fairly stable, but additional structure will reflect seasonality, and hence the planning horizon for a network is

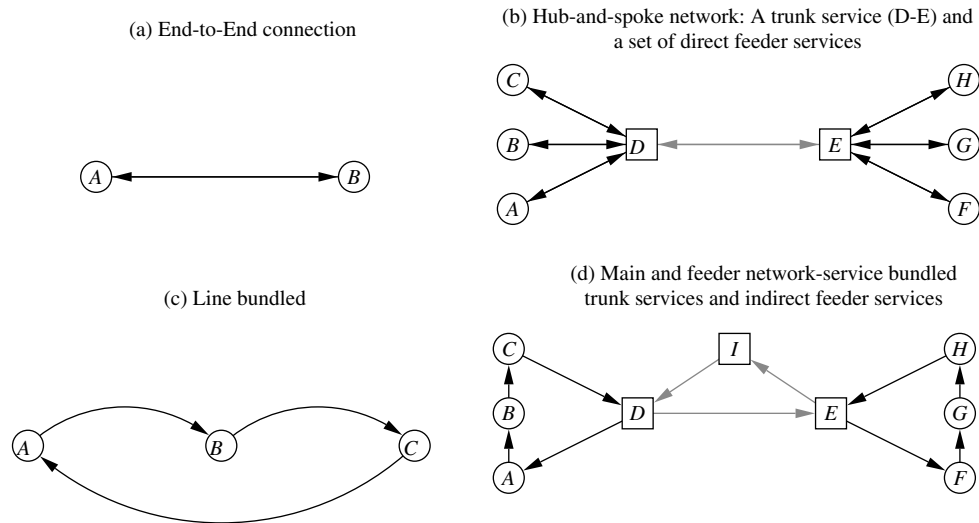


Figure 1 Examples of Routes and Network Designs

Note. Square nodes symbolize hubs or main ports, whereas round nodes symbolize feeder ports.

important to the liner service network design and to the fleet deployment. General economic and financial conditions have a major impact on the liner service network design and fleet deployment, but are hard to predict compared to a seasonal pattern, which may be recognized and accounted for.

4.2.3. Service. A service from the carriers' point of view consists of a set of port calls with designated arrival and departure times at a given frequency. The individual services are required to be good components of the network with regards to efficient transshipment facilities and fast direct services for critical connections. Notteboom (2004, 2006) categorizes a wide range of rotation patterns as follows.

End-to-End: A direct shuttle service between two ports (see Figure 1). *Line bundled:* A rotation visiting a set of ports in a loop. *Trunk services:* An end-to-end connection between hubs. *Direct feeder services:* Shuttle from feeder port directly to a hub. *Indirect feeder services:* A service-bundled rotation to a set of feeder ports and a hub. *Round-the-world (RTW):* A rotation following the equatorial belt visiting hubs to service the east-west and north-south trades in a grid. This type of service has a capacity constraint because it must traverse the Panama Canal restricted to panamax ships. *Pendulum:* A service traveling back and forth like a pendulum, e.g., Europe–Far East–U.S. West Coast–Far East–Europe. *Butterfly:* Multiple cycles centered around one port (Figure 2). Each cycle visiting an alternating sequence of ports. One cycle may be a subset of another. *Conveyor belt:* A service connecting regional hubs designed for transshipments between continents at the crossing point of trade lanes.

Rotation turnaround time varies from a single week up to 20 weeks, although the average rotation is

around 8–9 weeks. The rotation turnaround time is composed of voyage time at sea and the service time at ports used for piloting in and out of the port, berthing and loading/unloading cargo. The number of port calls in a rotation is a trade-off between economies of scale and transit times (Notteboom 2006). Visiting many ports leads to a good utilization of the vessel capacity at the cost of long transit times. Another complicating factor is that port stays are very time consuming. Notteboom (2006) reports that 21% of the transit time is the accumulated port stay for a China Ocean Shipping Company (COSCO) Europe–Far East service. The schedule includes buffer time to account for delays due to weather, port congestion, or unusually high terminal handling time. These delays may cause speed increases on individual voyage legs, which increases energy consumption and CO₂ emission.

4.2.4. Frequency. As described in §4.1, the liner-shipping business is characterized by having a public schedule. From the carrier's point of view, a schedule

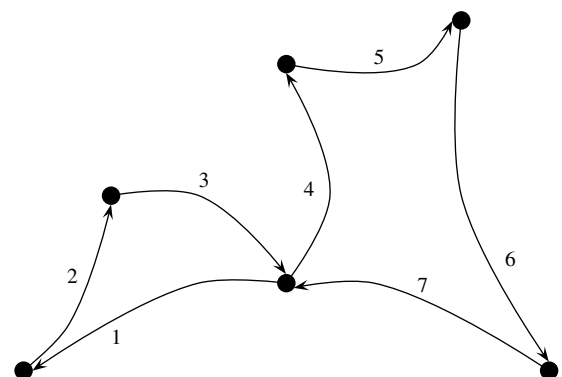


Figure 2 An Example of a Butterfly Rotation

consists of a set of *services*. The services consist of a fixed itinerary of ports typically called with weekly or biweekly frequency (Notteboom and Vernimmen 2009; Stopford 2009). The fixed (bi)weekly frequency is widely used in container liner shipping because of significant planning advantages for carriers, shippers, and terminals:

- *Reliability*. Fixed departures enabling complete integration of customer supply chains.
- *Simplified network planning*. A guiding rule for the carrier when designing the network.
- *Asset planning*. The use of port berths and vessels can be planned for better utilization.
- *Planning routing scenarios*. Synchronization of connecting services may provide timely transshipments for critical connections.

The stable flow of general cargo in containers enables the carrier to maintain a weekly or biweekly service. For instance, the weekly frequency of a schedule is achieved by deploying multiple vessels sailing one week apart, as illustrated in Figure 3. An example of two services connecting and transshipping at a hub port is shown in Figure 4. The speed of each voyage between port pairs is thus closely related to the number of vessels. Therefore, adjusting the number of vessels on a service is a means to adjust speed. At the same time, the decision of speed enables the carrier to adjust to demand fluctuation because the carrier can free up vessels by increasing speed. These vessels can then be deployed elsewhere to increase capacity.

4.2.5. Business Rules for Service Generation.

The generation of services has a high degree of freedom in terms of the number of port calls and average rotation time. In spite of the route flexibility, there are certain business restrictions arising from

complex real-world constraints. The real-world constraints may be expressed as “rules of thumb” to the combination of the vessel class deployed, the rotation turnaround time, and service frequency, according to Network and Product at Maersk Line (2011). Some important business rules are stated as follows.

1. Services deploying a vessel class of capacity of at least 1,200 forty feet equivalent unit (FFE) must have weekly frequency. This is a basic structure in *liner shipping*, because customers often plan their production for weekly deliveries. It also facilitates easy management of shared resources between carriers as terminals and canals. Smaller vessels calling smaller ports are exempt from this because of low demand and/or unstable terminal service. Vessels with capacity below 1,200 FFE are required to have at least biweekly frequency.
2. Services deploying a vessel class of capacity of at least 4,200 FFE must have at least a four-week rotation time, because larger vessels require higher volumes to fill and thus be economically competitive.
3. Services deploying a vessel class of capacity of at most 800 FFE must have at least two feeder port calls in between every hub-port call.

Business rules 1 and 2 are implemented in our algorithm and thus present in the computational results. Business rule 3 has not been implemented, as the proposed heuristic column generation algorithm does not contain the logic required for port call sequencing.

4.3. Assets and Infrastructure

4.3.1. Cost Structure and Economies of Scale.

The network cost of a carrier can be divided into fleet cost and cargo-handling cost, detailed as follows (we omit the administrative overhead estimated to around 30% by Stopford 2009):

• Fleet cost:

1. *Bunker cost* is the cost of bunker, which is the fuel consumed by container vessels.
2. *Capital cost* is the cost of acquiring or financing a single vessel v .
3. *Port call cost* is a terminal fee for calling a terminal with a given vessel v . (See next for costs related to cargo handling at ports.)
4. *Canal cost* is the cost of traversing a canal with a given vessel.
5. *Operational cost (OPEX)* is the operating cost of a vessel, including crew, maintenance, and insurance.

• Cargo-handling cost:

1. *(Un)load cost* is the container-handling cost, full or empty, at a given port.
2. *Transshipment cost* is the cost of transshipping a container, full or empty, in a port.
3. *Equipment cost* is the cost of owning or leasing containers.

According to Stopford (2009, Table 13.9), the bunker cost is 35%–50% of a vessel’s cost, capital cost is

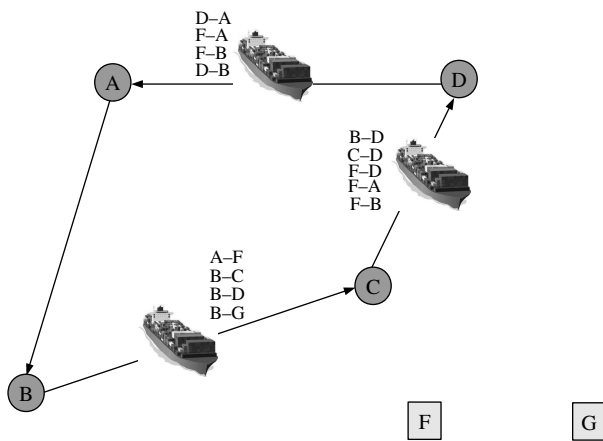


Figure 3 A Single Service Connecting Ports A, B, C, and D

Notes. The vessels are depicted along with a cargo list specifying the current cargo load. Total round trip time is 21 days and to provide weekly frequency three vessels are sailing one week apart. The cargo is destined for nodes on the service as well as nodes not belonging to the service. The cargo has multiple origins and destinations.

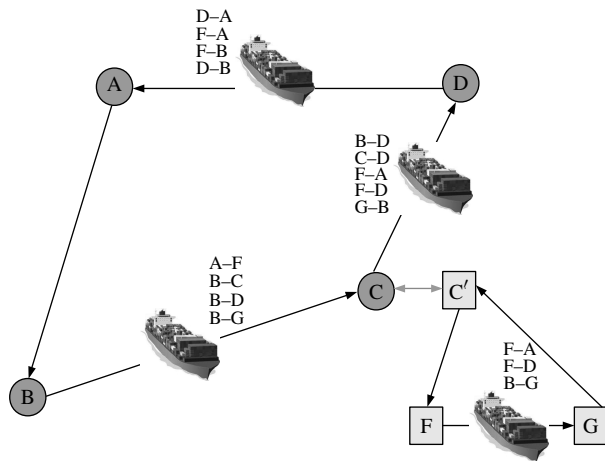


Figure 4 Two Connecting Services

Notes. The service from Figure 3 and a second service with a round trip time of one week illustrated by single vessels. The cargo composition on board vessels illustrate transshipments at the core of the liner-shipping network design.

30%–45%, OPEX is 6%–17%, and port cost is 9%–14%. Generally, bunker cost exceeds capital costs (apart from the largest vessels).

Bunker consumption depends on the vessel type, the speed of operation, the draft of the vessel (i.e., the actual load), the number of operational reefer containers powered by the vessel's engine, and the weather. Bunker consumption for a vessel profile is often approximated by a cubic function of speed per time unit (Alderton 2005; Stopford 2009). An empirical study of bunker consumption for container vessels at different speeds is presented in Wang and Meng (2012c), who conclude that a cubic function of speed is a good approximation of the bunker consumption. During a round trip, the vessel may sail at different speeds between ports. The vessel may *slow steam* to save bunker fuel or increase speed to meet a crucial transit time. Speed may be constrained by hard weather conditions or navigation through specific areas.

Both capital and operational cost varies with capacity. Economies of scale means that a large vessel is cheaper to operate per FFE, which is the most common container unit (Stopford 2009). The market rate of a vessel is called *time charter rate (TC rate)* and represents the cost of leasing (charter in) a container vessel into the fleet or for a carrier to forward lease (charter out) an owned vessel to another carrier. The TC rate fluctuates with seasonality and is highly dependent on the length of the chartering period. A carrier may have an owned fleet supplemented by chartering in and out to meet capacity requirements and to gain flexibility in asset management. TC rate will include daily running costs of the vessel, such as crew costs, repair, and maintenance.

For vessels owned by the carrier, the TC rate will cover operational and capital cost and depreciation of the vessel's value (see Stopford 2009, p. 544). It is assumed that the TC costs represent a market rate, where the carrier will be able to charter out the vessel in case of a surplus of vessels. The methods described herein will consider fixed TC costs, not considering financial asset management of a fleet of container vessels under an expected development of TC costs. For more details on such ideas, see Alvarez et al. (2011).

Port call cost and canal costs may be treated identically. The port call cost is a fee paid to the terminal. The fee depends on the size of the vessel, i.e., the capacity booked in the terminal and also on the geographical location and size of the terminal. The cost of traversing the Suez Canal and the Panama Canal depends on the size of the vessel and the actual load of the vessel.

(Un)loading and transshipment costs are also known as *cargo-handling costs*. The loading and unloading costs are fixed once the cargo is selected for transport, but the transshipment cost depends on the routing of the product, and hence the total number of transshipments. A global carrier will not provide direct connections for a significant percentage of the available transport scenarios, which incur transshipment at least once during a voyage. The cargo-handling cost can be a nonlinear function because some terminals have volume-dependent costs.

Equipment cost is the cost of containers. The carrier can own or charter the containers used for cargo transport (Stopford 2009). Stopford (2009) estimates the daily cost of an FFE dry container to around \$1 per day, whereas a reefer FFE has a daily cost of around \$5.60.

4.3.2. Vessels. An oceangoing container vessel is the core part of a carriers' operations. It can be characterized by specifications as *FFE capacity, weight capacity, speed, length, beam, draft, number of reefer plugs, ice class, age, engine power, etc.* The defining attribute is FFE capacity given as a nominal number. The actual capacity of a vessel depends on the service it is sailing and the actual cargo on board. A vessel cannot accommodate more reefer containers than it has reefer plugs. The draft, length, and beam of the vessel dictate which ports, canals, and straits the vessel can access. Some waters have special access restrictions like the Gulf of Finland, which during winter requires ice class vessels for service.

With regards to network design, the vessels are grouped in vessel classes with similar properties such as capacity and speed interval, e.g., a Panamax vessel class denoting the maximal width for traversal of the Panama Canal. Other common groupings are according to a capacity band, i.e., vessels with a nominal capacity of 1,500–2,100 twenty-foot equivalent units (TEUs).

Each vessel has a minimum speed $S_{\min}$, and a maximum speed $S_{\max}$, in knots, and a *design speed* S^* , and *design draft* at which its design fuel consumption F^* is optimized. Still, the actual speed largely decides the fuel consumption. Large vessels may use in excess of 200 metric tons (mt) of fuel per sailing day. Additional fuel is consumed by auxiliary engines for other vessel systems (1–12 mt/day) and for electricity for reefer units (a rule of thumb is 0.025 mt/day/reefer, depending on inside/outside temperature). Calculating actual fuel consumption is very complex because vessel draft, wind, waves, currents, and date of the last hull scraping affect fuel consumption.

Some seaboards are under sulphur emission restrictions, limiting the percentage of sulphur content in the bunker, denoted LSFO (low-sulphur fuel oil, as opposed to high-sulphur fuel oil (HSFO)). LSFO is supplied with a premium to the bunker price and has reduced availability; for details, refer to Plum and Jensen (2007).

4.3.3. Ports. A port may consist of several terminals competing for the cargo traffic in the corresponding port. A carrier will typically use a single terminal at every port because of connections between services. Ports have a maximum draft, and the berths have a maximum length. This results in port-vessel incompatibilities for some port-vessel combinations. A container vessel is piloted in and out of port by a pilot employed at the port authorities. *Pilot times* may be several hours, and for ports situated up a river bed it can be 8–10 hours. At some terminals a berthing slot is reserved in advance, whereas others serve vessels by a first-come, first-served (FCFS) basis. A vessel may have to wait for a given pilot time or wait for an available berthing slot at FCFS ports. To enter or leave some ports, a vessel may also have to wait for high tide. A port-calling fee, depending on the vessel's specifications, is levied by the port authority. The port-calling fee covers expenses to the pilots, tug boats, port authorities, and the terminal. Once the vessel has entered a terminal, the vessel will commence unloading and loading of cargo. Cargo handling is often referred to as *moves* in general. Each move is associated with a cost to cover the expenses of the cranes, terminal crew, and terminal administration. A transshipment move will typically cost less than a load and an unload move together. The crane height and length of the terminal may also result in port-vessel incompatibilities. The number of moves per unit time that a port is able to perform is denoted as its *productivity*. There is usually a distinction between main, transshipment, and feeder ports. Feeder ports are usually small, and their productivity may vary with the level of technology deployed at the feeder port. Main ports have a large quantity of import/export and some

transshipment facilities. Most main ports will have medium to high productivity. Lastly, transshipment ports such as Balboa, Singapore, and Algeciras do not have extensive import/exports, but serve to transship between services and intermodal transport. Transshipment ports usually have a high productivity.

4.3.4. Port Stay. In real life, port stay times will vary greatly because of the different productivity and strategies of terminals either having reserved berthing slots, or perhaps serving vessels on a FCFS basis. Furthermore, the port stay depends on the amount of cargo to load and unload in the given port for that particular calling. The benchmark suite does not include measures of productivity, nor do we model the port time on the basis of the cargo flow, and therefore it has been chosen to fix the port time to an identical measure for all ports and vessel classes. This is motivated by the observation that larger vessels, in practice, will have higher productivity, since they call on more efficient ports served by more cranes. This means that vessels, irrespective of size, often have comparable port stay times; an analysis of Maersk Line port stay times roughly supports this.

4.3.5. Canals. The two large canals of Panama and Suez enable fast transport between continents. To traverse the canals, a substantial fee is charged. The canals offer significantly faster transit times and also reduced operational costs because of the reduction of the sailing distance. Some sailing passages may have draft and width restrictions such as the Panama or Kieler Canal, resulting in a significant detour for larger vessels. Some, as the Bosphorus strait, are the only entrance to a sea body and the limitations of these straits thus dictate the size of entering vessels. The current restriction on vessel size of the Panama Canal is very decisive in the services of the American east coast from the Far East and Australasia. No such restriction exists for the Suez Canal with current container vessels.

4.3.6. Equipment. Containers are generally available in lengths of 20, 40, and 45 feet, each of which exist in different types as: dry, refrigerating (*reefers*), open top, and rack, plus additional specialized containers. A customer will generally require a specific type of container, although in some cases containers are used outside their designated scope. *Dry* equipment covers roughly 80% of a carrier's equipment pool (Stopford 2009), with the remainder mainly being *reefers*. There is a similar distribution on container sizes, with around 80% being 40-foot containers (Stopford 2009).

4.4. Demand and Customers

The goods and customers of containerized transports are plentiful and cover any type of manufactured

goods, but can also be bulk-like cargo such as stone, wastepaper, or refrigerated commodities like fish or fruits. A service will usually focus on customers in a *trade* (sometimes multiple trades), e.g., *Asia–Europe* or *South America–North America*. They may be grouped into east-west trades, which have larger volumes, allowing for economy of scale deploying huge vessels, and into north-south trades, which have much refrigerated cargo requiring specialized vessels and faster transit time. Similarly, all trades have special characteristics with regards to volume, service-level requirements, etc., that govern how a service can compete in the trade.

4.4.1. Demand. The production and consumption of some commodities vary over the year, some following harvest seasons, others following holidays and festivals or summer vacations. As a result, demand varies, some on a global level, others only affecting a single port or trade. The biggest of these is Christmas, which generates the yearly peak season in the third quarter, allowing warehouses to fill up prior to Christmas shopping. Another peak happens prior to Chinese New Year, because Chinese factories export their goods before closing down for the festivities. A customer or shipper will usually only pay for a container transport once it is on the vessel or delivered, i.e., there is no fee for booking and no fee for not submitting cargo for a booking. This risk-free booking policy for the shipper causes many problems for the carrier and shipper:

- It makes it hard to forecast the demand of cargo for some vessel departures (although it can be used at ports called later), so the carrier will overbook departures. This causes delays of cargo when forecasts and bookings fall short.

- Shippers know that overbooking is a risk, and to counter this, they will intentionally book for more containers than they expect to ship, often spread at different carriers to optimize flexibility. This especially happens in peak seasons such as the third quarter, where capacities are most pressed.

Containerized transport has been a growing industry for many years. Between 1983 and 2006 world gross domestic product (GDP) growth was 4.8% per year, whereas container cargo growth averaged 10% per year (Stopford 2009). The trend has continued in recent years, looking at the data from The World Bank (2010) with an exception of the years 2009 and 2011, where the industry was suffering from the worldwide financial crisis. The long history of fast growth in containerized cargo has given an expectation of continuous growth in the industry. The delivery time for new-build vessels is several years, and this makes the market slow to adapt to increased demand with resulting huge fluctuations in revenues, as seen in the aftermath of the financial crisis in 2009.

4.4.2. Revenue. In general, the revenue of a cargo is closely related to the *demand-supply* balance between the volume of cargo to be transported and the capacity offered by container carriers. However, many other aspects come into play. Specialized cargo, requiring specific equipment or different administrative handling, will give higher revenue, for instance, refrigerated or military cargo. Because the demand is not symmetrical (e.g., Asia exports more than is imported) but the supply is symmetrical (vessels return to Asia to be reloaded), hence, the revenue is also not symmetric. Transporting a container from Asia to Europe can easily cost three times more than the reverse (Alphaliner charter rates 2000–2010 2010).

The demand fluctuations over a year influence the revenues, with the highest revenues in the third quarter, but with variations over trades.

4.4.3. Service Level. The specific service a customer is presented with, for transporting a container from port *A* to port *B*, the *product*, can be classified by a number of parameters:

- The price: A basic rate, subject to additional fees and surcharges.

- Bunker adjustment factor (BAF): A variable price component dependent on the bunker price, making it possible for the carrier to share the risk of oil price volatility with the shipper.

- Transit time: The time to transport a container from *A* to *B*.

- Transshipments: The cargo is reloaded onto a new vessel in a third port. A direct product is usually preferred because of the decreased risk of missing a connection.

- Frequency: A weekly service calling a port is preferred to a biweekly service because of increased flexibility.

- Reliability: Ocean carriers differ in reliability of services and the products they operate (Notteboom 2006; Maersk Line 2011).

- Paperwork: As documentation and administration, different carriers and products require different paperwork, adding complexity for the shipper.

- Equipment: The carrier must supply the customer with a container prior to transporting.

- Weekday: Vessels departing on a Friday can, e.g., be preferable to a departure on Monday, because it can shorten the supply chain by three days (for a factory closed on the weekend).

All these factors can be relevant for a shipper, with price and transit time often being the key factors.

4.4.4. Routing (Products). As a result of trade policies, there may be various *cabotage rules* in countries restricting internal transport within the country to routings on vessels flagged/owned/operated in the country. Cabotage applies to transshipments

of cargo destined within the country and also for empty containers. Other special rules may exist, e.g., embargoes: when country *A* is embargoing country *B*, cargo from/to *B* cannot be transshipped in *A*. Dangerous goods are subject to *IMO* (International Maritime Organization) rules with regard to the quantity and the placement of these goods on the container vessel.

4.4.5. Competition and Partnerships. As mentioned, the economies of scale in liner shipping can generate huge savings for larger volumes. This motivates carriers to cooperate on operating services or subcontract. Such partnerships exist in various forms.

Foreign feeder agreements (FEF). A service operating between a hub and a few proximate ports is called a *feeder*. These are usually operated by smaller vessels (less than 1,250 FFE), and the cargo is transshipping in the hub to some oceangoing service. The feeder carrier will combine volumes from other global carriers to achieve economy of scale.

Alliances and vessel-sharing agreements (VSA). The significant economies of scale achieved by deploying larger vessels is the motivation for multiple carriers to enter a VSA. Two or more carriers share a round trip, making it possible to deploy larger vessels and provide higher frequency of service. Deploying large vessels presents a significant savings in operating cost for each carrier (see Stopford 2009, Figure 13.11) without reducing the service levels presented to their customers. In practice, VSAs are complicated because the partners seldom have matching services and demand, available vessels, etc., but VSAs are widely used and a central part of liner shipping. For more details on the mechanics of VSAs, refer to Agarwal and Ergun (2008a) for game-theoretic observations on the subject.

Slot charter agreements (SCA) are a combination of FEFs and VSAs. One carrier will enter a contract to use capacity (*slots*) on another carrier's service. This can either be on the full round trip of the service, allowing for the slot to be used differently in head and backhaul, or on a specified part of the rotation. The contract is usually for a fixed number of slots, which is paid for *used/unused*, sometimes with an option for extra slots as *pay per use*.

5. The Benchmark Suite for LSNBP

The purpose of the benchmark suite is to provide data for prototype implementations of models and methods for the liner-shipping network design and fleet deployment problem at a strategic level. The literature review displays three important factors for the development of the benchmark suite, as shown in the following:

1. The research community views the problem from different angles in terms of constraints and costs. As a consequence, the data must address the important factors seen across the entire research field

and also incorporate data on foreseeable extensions to current models.

2. A "base" model is needed to compare methods.

3. Currently, exact methods only solve very small instances, whereas heuristics such as the matheuristic by Alvarez (2009) caters for a large instance. As a consequence, the benchmark suite should contain small, realistic instances that may be solved to optimality within the foreseeable future and large instances for heuristic development.

There are at least two pressing research questions on the LSNBP: efficient modeling of bunker consumption as a cubic function of speed, and incorporating the level of service. Global carriers such as Maersk Line put great effort into creating a network design of low bunker consumption to meet a demand for environmentally friendly transportation of goods and improve the economy of the network. As noted by Notteboom and Vernimmen (2009), the price of bunker has a significant impact on the network design and the fleet deployed. Therefore, formulas for bunker consumption based on vessel class are included in the benchmark suite to allow for calculations of the overall bunker consumption as a function of speed. At the same time, Notteboom and Vernimmen (2009) state that customers are demanding a high frequency and low transit times, and therefore level of service and competitive transit times are important factors in LSNBP. The benchmark suite includes a set of maximal transit times on commodities as we believe level of service and transit time incorporation to be important open research questions in the liner service network design problem.

We have chosen to provide vessel data in terms of six fictitious, but realistic, vessel classes, where multiple vessels share similar characteristics. We believe a global carrier has several similar vessels assigned to a service (Agarwal and Ergun 2008b; Notteboom and Vernimmen 2009; Alvarez 2009). A prototype implementation will contain the most important details, but omit details believed to be nondecisive for the conceptual understanding of the problem. Therefore, the benchmark suite does not contain different equipment types such as reefer and high cube containers. The weight of vessels and cargo in particular are omitted from the benchmark instances.

Assigning specific vessels to a specific service is an operational issue dependent on cabotage rules. Hence, the benchmark instances do not indicate the country of registration of vessels in the present data set. Likewise, IMO rules apply to specific cargo and stowage plans, which is out of scope of the considered LSNBP.

Weather and currents may have a large influence on travel times and energy consumption in an LSNBP, but we have chosen to ignore these factors to avoid further complexity.

The benchmark instances are not meant for actual scheduling of services. Therefore, tidal information on ports is not provided, although they may be decisive for a specific berthing window at a port. On the same note, the specific weekday on which a demand becomes available for transport out of a given port is not included in the data of the benchmark suite.

The following section discusses data objects and data generation. A discussion on the range of instances needed to qualify the benchmark suite follows.

The benchmark suite originates from publicly available data and past data for Maersk Line.

External data sources are: “Distances between ports—Publication 151,” National Imagery and Mapping Agency (2001); The Hamburg Index, Vereinigung Hamburger Schiffsmakler und Schiffsagenten (2011); “Charter Rates 2000–2010,” Alphaliner charter rates 2000–2010 (2010); “Container Freight Rate Insight,” Drewry Shipping Consultants (2010).

The data from Maersk Line are from an undisclosed year and has been subject to reasonable random perturbation and measures to anonymize the data to protect the confidentiality of Maersk Line.

The purpose of the benchmark suite is to qualify algorithms for scalability, quality, and robustness by providing realistic data for the scale, structure, and complexity of the liner-shipping network design problem. We have chosen not to anonymize the actual ports to evaluate the network from a business perspective, but the data still do not represent an actual business case, nor does it contain all relevant data. We hope the benchmark suite can be used to perform strategic analysis on various scenarios. The

instances may be found at <http://www.linerlib.org> under the name *LINER-LIB 2012*. The following section describes the data generation process in terms of the origin of the data and the logic used to derive the benchmark suite.

5.1. Data Objects and Data Generation

An instance consists of the following data: *Port list* with port call cost as a function of vessel capacity (explained in more detail in §5.2); *fleet list* with design speed and bunker consumption at design speed and idling (§5.3); *distances* contains an all-to-all list of travel distances (§5.4); and *demands* specifies a set of origin-destination pairs and their corresponding demands (§5.5). Finally, the *graph* file defines a graph representation of the considered problem with way points. The data items are illustrated in Figure 5 giving an overview of the attributes of each item.

LINER-LIB 12 does not dictate the length of the planning horizon. Given that weekly service frequency is most often encountered in practice (Notteboom and Vernimmen 2009; Network and Product at Maersk Line 2011), the demand figures provided correspond to weekly volumes. One may obtain total flow volumes for an arbitrary planning horizon by scaling the weekly figures (e.g., multiply the given volumes by 52 to obtain yearly transport demand). The cost and revenues are not meant to reflect the actual cost of any carrier. The costs and revenues are constructed with care to reflect the relative cost structure within the network. This means that a large port is proportionally cheaper to call at than a small port. Likewise, it is cheaper to transship a container in Asia than in Europe or the United States.

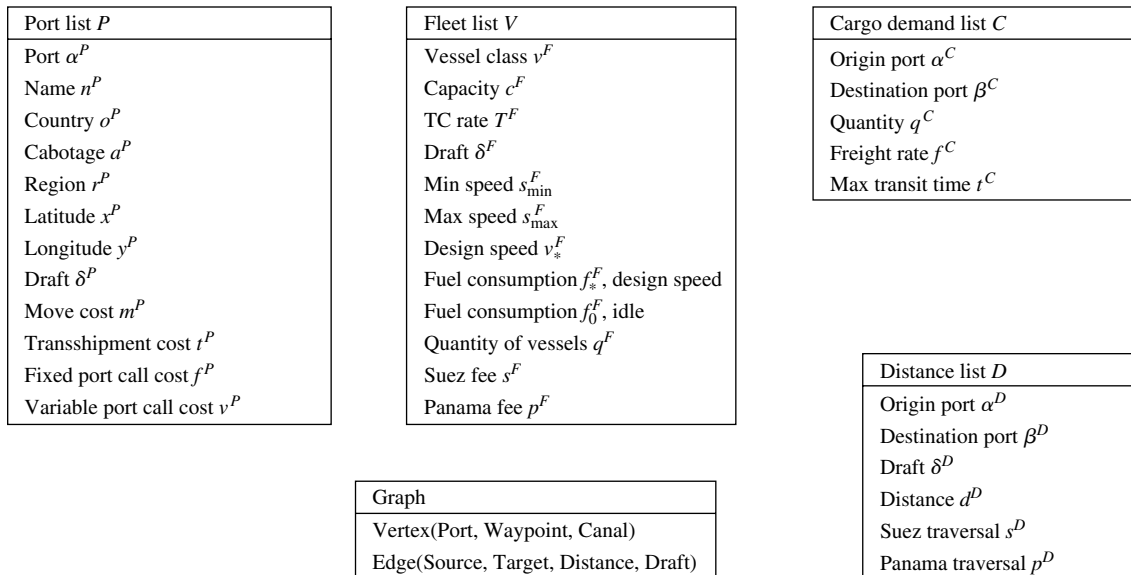


Figure 5 Data Objects of an Instance

Table 2 Example of Port Entry

Port α^P	Name n^P	Country o^P	Cabotage a^P	Region r^P	Long x^P	Lat y^P	Draft δ^P	Move m^P	Trans t^P	Fixed f^P	Var v^P
CNYTN	Shenzhen	China	China	South China	114.26	22.58	13.5	176.69	78.14	7,220.21	3.8

Notes. Port is given by UNLOCODE, e.g., CNYTN stands for the Chinese Port of Yantian. Draft is the maximal draft, Move is the (un)load cost, and Trans is the cost of a full transfer for transshipment. Fixed is the fixed port call cost and Var is the variable port call cost as a function of the capacity.

Revenues reflect demand and supply such that visiting a distant port with low demand requires higher revenues per FFE than a central port with extensive own demand and supply.

5.2. Port List

Ports are identified by UNLOCODE, a unique five-character identifier (for Europe 2011). The port list specifies the ports in the instance and contains the following fields: Port (in UNLOCODE); Name; Country; Continent; Cabotage region (if applicable); Revenue region (maps port to the revenue data supplied by Drewry Shipping Consultants 2010); Latitude; Longitude; Maximal berthing length (in meters); Maximal acceptable vessel draft (in meters); (Un)Load cost per FFE (forty foot equivalent unit) (in USD); Transshipment cost per FFE (in USD); Fixed port call cost (in USD); Variable port call cost per FFE (in USD), as seen in Table 2.

The port list reflects real ports ensuring that a solution can be mapped to a geographical coverage. This allows network planners and others without optimization background to evaluate a proposed network.

The transshipment cost will usually be lower than the sum of loading and discharging because transshipping does not require customs paperwork at the terminal in question. In an optimization model, the load and discharge may be viewed as fixed costs once the demand is chosen for transport.

The geographical location and the port size will determine a fixed cost for visiting the port and a variable cost related to the capacity of the vessel visiting the port. This relation is deduced experimentally from the perturbed Maersk Line data giving a

nonzero fixed cost and a variable cost corresponding approximately to a linear function of the capacity.

5.3. Fleet List

Six generalized vessel classes are constructed from the fleet list of Maersk Line (A. P. Møller-Mærsk Fleet List 2008) representing realistic capacity classes in the network. A vessel class will contain the following information: TC rate per day per vessel (in USD), this value includes operational costs; capacity (in FFE); design draft (in meters); minimal speed (in knots); maximal speed (in knots); design speed (in knots); daily bunker consumption (in metric tons) at design speed; own consumption (in metric tons) when idling at ports; number of vessels in the vessel class currently in the fleet; Suez canal fee (in USD); Panama canal fee (in USD), as seen in Table 3.

The minimal and maximal speeds referred to in the fleet list are not the technical minimum/maximum speeds. The speed interval is related to the average speed on a service deploying the vessel class in question. This means the vessel will be doing less than the minimal speed on part of the service, and hence the minimal speed referred to in the fleet list is higher than the technical minimum speed. The technical maximum speed will also be higher than the maximum speed referred to in the fleet list and likewise refers to the maximal average speed that can be assigned to a service with the vessel class in question deployed. The minimal speed is the “slow steaming” speed of a service deploying the vessel class. The only size measure describing the vessels is draft. Other measures such as length and beam width

Table 3 Example of the Fleet List

Vessel class v^F	Cap c^F	TC T^F	Draft δ^F	Min s $S_{\min}^F$	Max s $S_{\max}^F$	Des s v_*^F	Fuel* f_*^F	Fuel0 f_0^F	Quantity q^F	Suez s^F	Panama p^F
Feeder_450	450	5,000	8.0	10	14	12.0	18.8	2.4	38	175,769	64,800
Feeder_800	800	8,000	9.5	10	17	14.0	23.7	2.5	77	218,445	115,200
Panamax_1200	1,200	11,000	12.0	12	19	18.0	52.5	4.0	124	267,217	172,800
Panamax_2400	2,400	21,000	11.0	12	22	16.0	57.4	5.3	161	413,533	345,600
Post_panamax	4,200	35,000	13.0	12	23	16.5	82.2	7.4	91	633,007	
Super_panamax	7,500	55,000	12.5	12	22	17.0	126.9	10.0	10	1,035,376	

Notes. Vessel class is the name of the class. Cap is the capacity in FFE. TC is the TC rate. Draft is a single measure expressing width, length, and draft for port/vessel compatibility. Min s , Max s is the minimum and maximum average speed interval of the vessel, whereas Des s is the design speed. Fuel*, Fuel0 are bunker consumption at design speed and idling, respectively. Quantity is the available number of vessels in the fleet with these specifications. Suez, Panama are the canal fees for the respective vessel classes.

should be seen as a proportional function of draft. This generalization is done to simplify the problem, keeping in mind that a full description of the problem can extend to more than this one size measure without losing applicability.

The fleet list reflects the demand scenario of the instance because a trade lane has a smaller fleet than a world instance.

5.3.1. TC Rates. The TC rates fluctuate with the market, and hence capturing TC rate is very dependent on the date it was retrieved. We have compared data from Alphaliner charter rates 2000–2010 (2010) and “HAMBURG INDEX Containership Time-Charter Rates” Vereinigung Hamburger Schiffsmakler und Schiffsagenten (2011), which have charter rates for vessels up to 4,000 and 4,800 TEU, respectively. For vessel classes above 4,800 TEU, we have constructed a TC rate based on the percentage increase of TC rate in the Maersk Line data to corresponding capacity intervals of the fictitious vessel classes and have increased the Hamburg index of charter prices with the corresponding percentage. The TC rates have been rounded up to the nearest thousand for vessel classes below post panamax size and to the nearest 5,000 for the two largest vessel classes.

5.3.2. Bunker Consumption. Bunker cost is a significant cost component of the network. The bunker curves for the generalized vessel classes are based on the formulas of Stopford (2009) and Alderton (2005). There is general agreement that bunker consumption F may be estimated by a cubic function

$$F(s) = (s/v_*^F)^3 \cdot f_*^F \quad (1)$$

for any speed s between the *min speed* $s_{\min}^F$ and *max speed* $s_{\max}^F$ of the vessel F , where v_*^F is the design speed, and f_*^F is the fuel consumption at design speed. This function disregards draft, although it is a significant factor, so the bunker consumption should be the consumption at design draft as well. The design draft is the draft the vessel has with an average load and for which it is designed to have the lowest bunker consumption as a function of speed. The fleet list specifies the design speed and consumption at design speed for the cubic formula for each vessel class. The bunker price is variable and the effect of bunker price could be a scenario for a user of LINER-LIB 12. In the computational results we use a flat bunker price of 600 USD per ton.

5.4. Distances

The distances between ports are based on information from the National Imagery and Mapping Agency (NIMA 2001). The data from NIMA contain distances from each port to major way points (ocean junctions) and ports in the vicinity. The data enable a mapping of global sailing routes onto a graph of ports and

Table 4 Example of Distance File

Origin α^D	Destination β^D	Distance d^D	Draft limit δ^D	Suez traversal s^D	Panama traversal p^D
GBABD	USOAK	7,953	12	No	Yes
GBABD	USOAK	13,862		No	No

Notes. Origin is the origin (in UNLOCODE), Destination is the destination (in UNLOCODE). Draft limit is the draft limit in meters (if relevant). Suez traversal, Panama traversal indicate whether the respective canals are passed to calculate canal fees.

way points. Each arc is given specifications such as the maximal draft and maximal width, such as the Panama canal, which has a width restriction. A single-source shortest-path algorithm between each pair of ports determines the shortest feasible path for each vessel class based on this data. This means that the distance between Oakland on the U.S. West Coast and Savannah on the U.S. East Coast will differ for vessels able to traverse the Panama Canal, as opposed to vessels not able to traverse the Panama Canal. To account for canal cost, a parameter indicates whether the distance is based on a visit to the Suez Canal and/or the Panama Canal. The distance file is a table with multiple entries for each port pair depending on draft limitations and canal usage.

The distance data consists of: Origin port α (in UNLOCODE); destination port β (in UNLOCODE); distance (in nautical miles); draft limit (in meters) δ^D ; Suez traversal (yes/no); Panama traversal (yes/no), as seen in Table 4.

The canal dues for the world’s two most important canals, Panama and Suez, are included in LINER-LIB 12. These are based on their published cost structure (Suez 2011; Panama Canal 2011) and created for the relevant vessels classes. Additional canals are omitted from LINER-LIB 12.

After manual inspection of the distances, several entries were identified to be erroneous. We have attempted to verify the correctness of several distances through external data sources and to correct obvious errors in the data. However, the distances are not guaranteed to be the shortest distances because of the selection of way points and possible undetected errors.

The distances have been collected and put into a graph $G = (V, E)$. The set of ports P and the set of way points W constitute $V = P \cup W$. The edge set E is from National Imagery and Mapping Agency (2001). Two distance files are generated.

1. An all-to-all distance matrix only for the port set $P \times P$. This graph has few vertexes and many edges.

2. A (sparse) graph $G = (V, E)$, of the instance with $V = P \cup W$ nodes.

The graph file contains all vertices and edges needed to navigate between any pair of ports in P with the fleet F . The files are generated using a shortest-path

Table 5 Example of Demand File

Origin α^c	Destination β^c	Quantity q^c	Freight rate f^c	Max. transit time t^c
USLSA	CNYAT	370	1,500	15

Notes. Origin, Destination give the origin and destination of the demand. Quantity states the number of FFE needed. Freight rate is the freight rate in USD per FFE. Max transit time states the maximum transit time in days.

algorithm between all pairs of ports for each vessel class in the fleet F .

5.5. Cargo Demands

Realistic demand data that capture the asymmetry in world trade is important when deciding on capacity and port sequences. A maximal transit time for each demand is provided for future models incorporating level of service or maximal transit time constraints.

The demand table contains the following information: UNLOCODE of the origin port α ; UNLOCODE of the destination port β ; quantity (in FFE); freight rate (in USD per FFE); maximal transit time (in days), as seen in Table 5. The demands are assumed to be the expected weekly demand.

For world instances, demands are aggregated onto the main ports.

The revenues are based on the trade lane prices found in “the container freight rate index,” courtesy of Drewry Shipping Consultants (2010). The rates are independent of any carrier and contains reliable market data collected across several carriers and markets. Drewry Shipping Consultants (2010) do not keep data for the South America west coast and some region pairs. In such cases, the official rates from the website of Hapag-Lloyd (2011) have been applied. The revenues are taken from a specific period of time, and it should be noted that these fluctuate greatly dependent on volume changes, new buildings, bunker price, etc. The revenue is 70% of the market freight rates because we are only dealing with the network cost and must be able to pay for the administrative overhead estimated to 30% by Stopford (2009). The volume of the origin-destination ports along with the distance to the nearest main or transshipment port are used to vary the trade lane price according to the demand structure, such that a demand from a transshipment or main port is cheaper than a demand from a feeder port, and feeder port revenues are dependent on the distance from the main port. Lastly, the freight rates are perturbed within an interval of $\pm 5\%$ to reflect price variance of different markets and cargo. The maximal transit time is set 30% higher than the fastest possible in Maersk Line’s network in 2010.

5.6. Instance Range of LINER-LIB 12

The instances are meant to challenge both exact and heuristic methods. Exact methods are currently

Table 6 The Instances of LINER-LIB 12 with Indication of the Number of Ports and the Number of Distinct Origin-Destination Pairs

Category	Instance and description	Ports	Dmnds
Single-hub instances	<i>Baltic</i> : Baltic sea with Bremerhaven as hub	12	22
	<i>WAF</i> : West Africa with Algeiras as hub to West African ports	19	38
Multihub instances	<i>Mediterranean</i> : Mediterranean with Algeiras, Tangier, Gioia Tauro as hubs	39	369
Trade lane instances	<i>Pacific</i> : Asia-U.S. West	45	722
	<i>AsiaEurope</i> : Europe, Middle East, and Far East regions	111	4,000
World instances	<i>Small</i> : 47 main ports worldwide identified by Maersk Line	47	1,764
	<i>Large</i> : 197 ports—the majority of ports serviced directly by Maersk Line	197	9,630

Note. The instances may be found at <http://www.or.man.dtu.dk/English/research/instances>.

limited to approximately 10 ports, whereas the tabu search heuristic of Alvarez (2009) provides results for 120 ports. Furthermore, the instances are meant to encourage the progress of method development and hence include instances of real-life scale, although we cannot yet solve them to optimality. Most of the instances are subsets of a larger network, where the subset is selected to highlight various aspects of the LSNDP, such as using a single hub or multiple hubs, the importance of draft restrictions or the capacity in terms of the fleet size is low compared to the demand, and vice versa. The provided instances are listed in Table 6. In addition to the instances, we have generated a set of rules to transform the *base* cases into two additional scenarios, where there is *low capacity* and *high capacity* compared to demand volume. The scenarios are created by adjusting the fleet data in terms of the number of vessels available and the TC rate, which is adjusted to reflect whether there is a deficit or surplus of capacity in the market. For the *low-capacity* case, TC rates are multiplied by 1.4 to reflect a 40% increase of TC rates, whereas the fleet quantity is decreased by 20% by multiplying the quantity with 0.8 and rounding to nearest integer value. In the *high-capacity* case, TC rates are decreased by 20% by multiplying TC rate by 0.8, and the fleet quantities are increased by 20% multiplying each vessel class quantity by 1.2 and rounding it to the nearest integer.

The Pacific and the single/multiple-hub instances should be challenging for the exact approaches and will also be applicable to model development because models of LSNDP can be verified by analyzing the results of LINER-LIB 12.

To have a medium-sized multihub instance, the Mediterranean case has been constructed. It includes 39 ports between Morocco and Suez, and includes ports in the Black Sea. Several ports have large demand sets throughout the seaboard. The demands

in the Mediterranean instances are constructed and not comparable to a real-world setting. The AsiaEurope trade lane instance and world instances should be challenging for the heuristics developing new methods, neighborhoods, and competing for the current best-known objective value. Exact methods solving these instances to optimality are not expected in the near future. We do not provide computational results for the world large instance because the size is currently considered out of reach even for heuristic methods.

LINER-LIB 12 can also be used in other optimization models related to liner-shipping network design, such as evaluation of a set of candidate services, synchronization of services to reduce transit times, sailing speed optimization, etc.

6. Reference Model for the LSNBP

The following model is based on the model presented in Alvarez (2009), with extensions to handle butterfly rotations and weekly or biweekly frequencies. Butterfly rotations (illustrated in Figure 6) make it more difficult to account for transshipment costs. The planning horizon remains open, but in the route generation a constraint has been added such that the number of vessels are aligned with the planning horizon and service duration to provide (bi)weekly frequency.

6.1. MIP Formulation

In this section, we present a possible formulation of the LSNBP. In the maritime sector, a service is defined as a sequence of ports to be visited following a published schedule. Whereas we adhere to this convention, the core entity of our model must be more specific to capture all transshipment costs properly. We therefore define a *rotation* as a particular

configuration of a service, vessel class, number of vessels deployed, and speed.

Our formulation of LSNBP addresses butterfly routes, because these are often used in practice. This means that the flow balance equations need to be extended. To balance the flow at butterfly nodes correctly, a three-index formulation is necessary. Consider the rotation in Figure 6, where port c is the butterfly port. With a traditional flow balance at port c , cargo that arrives from b might be redirected toward port a under the same conditions as cargo that continues on to port d . However, cargo travelling on the port sequence $b-c-a$ must be unloaded at c as the vessel continues to d , and later reloaded as the vessel arrives from e . Hence, it appears that we must track the latest port visited by the cargo, as well as its immediate destination, which we can encapsulate using a three-index formulation. Notice that the cardinality of the problem is not affected by this representation, given that one requires the same number of port pairs or port triples to specify a service uniquely.

The following sets are used in the formulation. (The square brackets describe how the sets relate to the graph defined in §5.4 and data objects defined in Figure 5):

- R** All rotations in the model, indexed using r .
- P** All ports in the model [$P := P$].
- E** Set of all possible edges in the model [$E \subset P \times P$]. All edges are directed and uncapacitated.
- E_r** Set of edges used in rotation r .
- Ω_r** Set of ordered port triples (h, i, j) in rotation r . The triples are ordered in the same manner as they will be visited by a vessel in the rotation.
- V** Vessel classes in the model, indexed using v [$V := F$].
- G** Set of port pairs with demand for transport, indexed using (o, d) [$G \subset P \times P$].

The following parameters represent the known data for the problem (square brackets describe how the parameters are calculated from LINER-LIB 12 data):

- a_{ij}^v Canal costs for vessels of type v when traversing arc (i, j) [s^F (Suez), p^F (Panama)].
- c^v Capacity (in FFE) of vessels of type v [c^F].
- d_j^v Port call costs for vessels of type v when entering port j [$f^P + v^P \times c^v$, where c^v is the number of FFEs the vessel can carry].
- e Fuel price per ton [\$600 USD for all instances].
- f^v Daily running costs for vessels of type v over the entire planning horizon [$f^v = T^F \cdot 180$ because we have a 180-day planning horizon, and daily TC rate is T^F].
- $\tilde{f}^v$ Cost or revenue of excluding a vessel of type v from operations. This can be the revenue obtained from chartering the vessel out over the

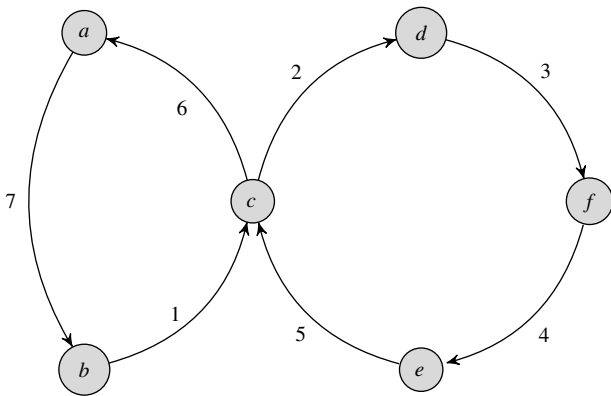


Figure 6 An Illustration of the Flow Balancing of Butterfly Routes

Notes. The numbers on the arcs denote the sequence the vessel sails in the butterfly. Commodities travelling from b to a either follows the vessel on route $b-c-d-f-e-c-a$ or are transshipped at c , where the handling cost of the container must be paid in full.

entire planning horizon, or the cost of laying up the vessel, depending on transportation market conditions [f^v (vessel in use), $f^v - \tilde{f}^v = 0$ (vessel not in use)].

g^{vs} **Fuel consumption** (tons per mile) for vessels of type v steaming at speed s [a table of possible values of speed, and g^{vs} is generated a priori by the program using Equation (1)].

h^v Fuel consumption (tons per day) for vessels of type v when idle at port [f_0^F].

k_{od} Total demand (in FFE) to the liner company for transport from o to d over the planning period [q^C where $o = \alpha^C$ and $d = \beta^C$].

l_{ij}^v Length in nautical miles of direct sailing from port i to port j using vessel type v . Notice, for instance, that canal width restrictions will result in different voyage lengths for wider vessels [d^D where $i = \alpha^D$, $j = \beta^D$, and $\delta^D > \delta^F$].

m_r Number of round trips performed by a vessel on rotation r during the planning period. We do not impose an integrality restriction on this parameter, because we assume that a fractional portion of the voyage can be completed at the beginning of the following planning period.

p_j^v Time at port j for vessels of type v [based on historical data, we set $p_j^v = 24$ h regardless of vessel size].

q_{od} Revenue from transport of one FFE from o to d [f^C where $o = \alpha^C$ and $d = \beta^C$].

$\tilde{q}_{od}$ **Penalty** for failing to transport one FFE from o to d [\$1,000 USD per FFE in all cases].

s_r **Speed** of all vessels used in rotation r [generated by the algorithm. The code tries several speed-vessel-route combinations (respecting the vessel speed range) and picks the one that leads to a better objective value for that iteration].

u_j Cost of lifting one FFE at port j [m^P].

t_j Cost of transshipping one FFE at port j [t^P].

v_r Vessel type used in rotation r .

z^v Number of vessels of type v available [q^F].

Finally, the decision variables used in the MIP are:

$X_{(ij)d}^r$ Number of containers travelling to their final destination at port d along edge (i, j) on rotation r .

$U_{(hi)d}^{rs}$ Number of containers travelling to port d that arrive to port i via edge (h, i) of rotation r for transshipment to rotation s .

$W_{(id)}^r$ Number of containers travelling to port d reaching their final destination via edge (i, d) using rotation r .

V_{od}^r Demand from port o to port d that enters the network for the first time because it is loaded to a vessel in rotation r .

O_{od} Demand from port o to port d that will not be serviced by the liner company.

Y^r Number of vessels assigned to rotation r .

The MIP model is presented next, followed by an explanation of the objective function and all the constraints.

(VRD) minimize Z_{MP}

$$= \sum_{r \in \mathbf{R}} f_{v_r}^r Y^r + \sum_{v \in \mathbf{V}} \tilde{f}^v \left(z^v - \sum_{r \in \mathbf{R}: v_r=v} Y^r \right) \quad (2a)$$

$$+ \sum_{r \in \mathbf{R}} m_r Y^r \sum_{(i,j) \in \mathbf{E}_r} (e h_{v_r}^r p_j^{v_r} + e g_{v_r, s_r}^{v_r} l_{ij}^{v_r} + d_{ij}^{v_r} + d_{ij}^{v_r}) \quad (2b)$$

$$+ \sum_{(o,d) \in \mathbf{G}} \left(\tilde{q}_{od} O_{od} - q_{od} \sum_{r \in \mathbf{R}} V_{od}^r \right) \quad (2c)$$

$$+ \sum_{r \in \mathbf{R}} \sum_{(h,i,j) \in \Omega_r} \left(u_i \left(W_{(hi)}^r + \sum_{\substack{d \in \mathbf{P} \\ d \neq i}} V_{id}^r \right) + t_i \sum_{\substack{d \in \mathbf{P} \\ s \in \mathbf{R} \\ d \neq i, s \neq r}} U_{(hi)d}^{rs} \right) \quad (2d)$$

$$\text{subject to } X_{(hi)d}^r + V_{id}^r + \sum_{\substack{s \in \mathbf{R}: \\ (k,i,l) \in \Omega_s \\ s \neq r}} U_{(kl)d}^{sr} = X_{(ij)d}^r + \sum_{\substack{s \in \mathbf{R} \\ s \neq r}} U_{(hi)d}^{rs}$$

$$r \in \mathbf{R}, (h, i, j) \in \Omega_r, d \in \mathbf{P} i \neq dh \neq d, \quad (3)$$

$$X_{(ij)d}^r = W_{(ij)d}^r \quad (i, j) \in \mathbf{E}_r, r \in \mathbf{R}, \quad (4)$$

$$O_{od} + \sum_{r \in \mathbf{R}} V_{od}^r = k_{od} \quad o, d \in \mathbf{G}, \quad (5)$$

$$\sum_{d \in \mathbf{P}} X_{ijd}^r \leq c_{v_r}^r \cdot m_r \cdot Y^r \quad r \in \mathbf{R} (i, j) \in \mathbf{E}_r, \quad (6)$$

$$\sum_{r \in \mathbf{R}: v_r=v} Y^r \leq z^v \quad v \in \mathbf{V}, \quad (7)$$

$$X_{(ij)d}^r, U_{(ij)d}^{rs}, W_{id}^r \in \mathbb{R}^+ \quad r, s \in \mathbf{R}, r \neq s \\ (i, j) \in \mathbf{E}_r, d \in \mathbf{P}, \quad (8)$$

$$O_{od}, V_{od}^r \in \mathbb{R}^+ \quad r \in \mathbf{R} o, d \in \mathbf{G}, \quad (9)$$

$$Y^r \in \mathbb{Z}^+ \quad r \in \mathbf{R}. \quad (10)$$

Expression (2a) in the objective function captures the daily running costs incurred by vessels in operation, as well as costs or revenues obtained when excluding some vessels from the operating fleet. The next term (2b) obtains the fuel costs (including consumption during steaming and while idling at ports), canal cost, and port-calling fees for all active vessels. The term (2c) obtains the total revenues from transporting cargo as requested, and the penalties incurred when rejecting carriage requests. Finally, the term (2d) computes charges from loading and unloading containers, both at their origin and destination, and at transshipment points.

Constraints (3) balance the flow of containers loaded and unloaded from each rotation at nodes

other than their final destination. Every container not destined for the port in question must either continue along in the same rotation or be unloaded for transshipment to a different rotation. Constraints (4) represent the flow of containers that were accepted for carriage arriving at their final destination. Every container arriving at its final destination is unloaded and grounded.

Constraints (5) tally demand from the originating hinterland. The variables O_{od} allow us to obtain a feasible solution for any fleet configuration. When there is not a sufficient number of vessels to transport all the demand, or when it is not economically convenient to transport all containers, the variables O_{od} are set to a positive value, indicating that some demand has been forfeited.

Constraints (6) impose restrictions on the total number of containers that can be transported by each edge in a rotation. The total that can be transported is limited by the number of vessels assigned to the rotation, the capacity of those vessels, and the number of trips the vessels perform over the planning horizon. Constraints (7) ensure that the number of vessels that are deployed does not exceed the number of available vessels of each type. Finally, constraints (8)–(10) impose nonnegativity and integrality restrictions on the corresponding variables.

6.1.1. Summary of Simplifications in the Model.

In this section we summarize some of the practical constraints and business rules stated in §4 that are *not* considered in the presented reference model for LSNBP. A subset of these constraints are accounted for in the data of the benchmark suite, whereas others are considered out of scope for mathematical models on the LSNBP. The reference model does not consider a maximal *transit time* for each commodity, and transit times are supplied for each commodity for future model development. Repositioning of empty containers are not dealt with, because real-world networks will usually require the flow of containers to balance. *Nonlinear handling costs* in ports are not considered in the reference model or the LINER-LIB 12 data. The *port productivity* and *pilot times* for berthing are not considered in the fixed port stay of the reference model. The reference model considers a single container type, and thus reefer capacity and the *bunker consumption of reefer containers* are not included in the reference model. Lastly, we do not consider *equipment cost*, *embargo*, or *cabotage rules*.

6.2. Metaheuristic

The heuristic that coordinates the overall algorithm is based on a heuristic column generation framework (not to be confused with the column generation in Desrosiers and Lübbecke 2005 in LP) where an auxiliary problem is used to generate new rotations

(columns) to the overall model (2)–(10), and a tabu search is applied to select the most promising set of rotations to constitute the network. At any iteration u , we solve a multicommodity flow problem (MCFP) with aggregated origins using the then-current set of rotations $\mathbf{R}^u$. The MCFP is defined in the $X_{(ij)d}^r$ variables for the edges of the rotation, the $U_{(hi)d}^{rs}$ variables to account for transshipments between rotations with edge cost t_i and $W_{(id)}^r, V_{id}^r$ variables for (un)load to ports at their origin or destination with edge costs u_i . Rejected demand (O_{od}) is modeled by adding an auxiliary edge (o, d) with cost $\tilde{q}_{od}$ for each commodity. To explain the transformation to the MCFP, we transform the current rotations and their transshipment, (un)loading possibilities into a graph $G = (N, A)$. We define the set of port terminal vertices N_T for every port $p \in P$, and we define a set of port call vertices for every rotation as N_C^r . For every vertex we define a function $p(u)$ that returns the port $p \in P$ represented by vertex u . The unified set of vertices is defined as $N = N_T \cup N_C^r$. The set of edges consist of load edges A_l (defined in the variables V_{id}^r with cost u_i and capacity c^{v_r}) connecting $w \in N_T$ to $v \in N_C^r$, where $p(w) = p(v)$. Conversely, unload edges A_u (defined in the variables $w_{(ij)}^r$ with cost u_j and capacity c^{v_r}) connect $w \in N_C^r$ to $v \in N_T$, where $p(w) = p(v)$. The set of transshipment edges A_t (defined in the variables $U_{(hi)d}^{rs}$) connect $w \in N_C^r$ to $v \in N_C^s$, where $p(w) = p(v)$ (for butterfly routes $r = s$), with cost t_i and capacity $\min\{c^{v_r}, c^{v_s}\}$. Define the set of sailing edges A_s (represented by the variables $X_{(ij)d}^r$), with zero cost (because the cost is entailed in the cost of the routing), and capacity c^{v_r} connecting every port pair $w, v \in N_C^r$, where w is the port call preceding v in the rotation defined by the solution. Finally, we represent rejecting demand by omission edges A_o (defined in the variables O_{od}) is defined for each commodity $g \in G$ with cost $\tilde{q}_{od}$ and capacity k_{od} . Define edge set $A = A_u \cup A_l \cup A_t \cup A_s \cup A_o$, and $G = (N, A)$ is a capacitated, directed graph for which we can solve a multicommodity flow problem defined with the quantities and revenues of the commodities in the set G from the original problem. The revenue is linked to the V_{od}^r variables for each commodity, and the objective function minimizes the cost of the edges subtracting the revenue of each commodity. The graph is illustrated in Figure 7.

The heuristic column generation scheme applies a tabu search on a set of rotations iteratively generated by a MIP neighborhood and is inspired by the approach presented in Alvarez (2009). The algorithm presented in this paper has a more advanced MIP neighborhood for generating rotations, which includes more heuristic measures on cargo composition and is designed to generate simple and butterfly routes of high quality. The MIP used for route

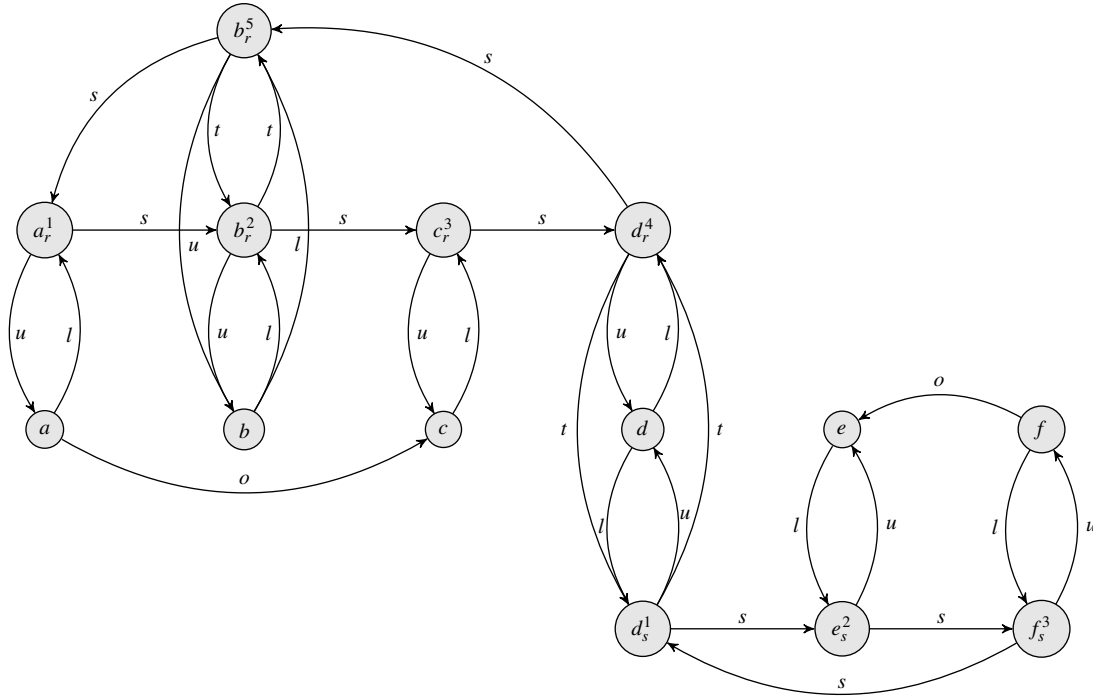


Figure 7 An Illustration of the Multicommodity Flow Graph of a Given Solution with Separation of Terminal Vertices and Port Call Vertices for Each Rotation

Notes. Terminal vertices are denoted by a letter designating a given port, whereas a port call vertex is designated by the port it represents, the route it is on, and the call number in the rotation. The graph is connected by (un)load edges (indicated by letters u and l), transshipment edges (t), sailing edges (s), and the penalty for rejecting cargo is modelled in the omission edges (o) between port pairs.

generation includes (bi)weekly frequency constraints and respects business rules 1 and 2 from §4.2.5.

The traffic demand that is not assigned to any of the routes in $\mathbf{R}^u$ is the residual traffic demand for the following iteration, $\hat{k}^{u+1}$. For $u = 0$ there are no rotations in $\mathbf{R}^u$, and $\hat{k}_{od}^0 = k_{od}$.

Let us assume that, for each vessel family v , a number n_v of vessels are available at the end of iteration u . For each family of vessels where $n_v > 0$, we launch a collection of subproblems $\text{AUX}(v, s, \kappa)$, for $s \in [s_{\min}^F, s_{\min}^F + 1, \dots, s_{\max}^F]$ and various values of the number of vessels deployed κ . In practice, this might give rise to a fairly large number of subproblems. Therefore, during the initial iterations, we focus on a narrow subset of speeds and number of vessels to be deployed.

In solving the subproblems $\text{AUX}(v, s, \kappa)$, we will typically obtain multiple feasible integer solutions. All these solutions are added to the overall model (2)–(10), since in any case the subproblem contains some approximations, and the optimal solution to the $\text{AUX}(v, s, \kappa)$ may not be the solution that advances the overall heuristic the most. Our current implementation parallelizes the subproblems, and we obtain batches of 16 subproblems that run simultaneously.

After the heuristic column generation for iteration u has terminated, we proceed to evaluate each new

column ζ_j in conjunction with $\mathbf{R}^u$. We set $\mathbf{R}_j^{u+1} = \mathbf{R}^u \cup \{\zeta_j\}$ and run the multicommodity flow algorithm on this network. We select from among the $\mathbf{R}_j^{u+1}$, the network with the best objective function, and set $\mathbf{R}^{u+1}$ to be this network.

At some iterations, none of the candidate networks will result in a globally improved objective function. We permit the algorithm to exploit this neighborhood for a limited number of iterations. If the overall objective function does not improve after a predefined number of iterations, we trigger a backtracking process. First, we return to the best-known configuration $\mathbf{R}^*$. Then we delete rotations from $\mathbf{R}^*$ based on three criteria: empty FFE miles, underutilized legs, and mixed-integer programming gap when the column was generated. The number of rotations that are deleted depend on how many times we have returned to this same $\mathbf{R}^*$ —a larger number of returns to $\mathbf{R}^*$ possibly indicates that this is no longer a promising point of departure, and the algorithm will remove a larger subset of rotations from it. The set of routes that results after backtracking is used as the basis for the following iteration.

If at any iteration the heuristic column generation procedure fails to produce any new columns (for instance, because no vessels are available), we delete a small number of rotations from $\mathbf{R}^u$. To reduce cycling,

we add recently added rotations to the tabu list to avoid it being deleted from $\mathbf{R}^u$.

6.3. Route Generation

The auxiliary problem $AUX(v, s, \kappa)$ generates rotations based on the current network. We allow one butterfly node per rotation. We also designate one port as the “master” port of the service for ease of writing the subtour elimination constraints.

The following additional parameters are required for the formulation of the auxiliary problem.

- $\hat{k}_{od}$ Residual demand (in FFE) to the liner company for transport from o to d . At any iteration in the heuristic, the residual demand is computed by taking the original demand and subtracting the flow that is carried by existing rotations.
- T Length of the planning horizon, in days.
- δ Empirical parameter, estimates the amount of additional flow flowing through a butterfly node, as compared to a regular node.
- $\phi_n^{\text{in}}, \phi_n^{\text{out}}$ Empirical parameters that capture the importance of a port as an exporter or importer.
- κ Number of sister vessels on the new service.
- v Vessel type to be deployed.
- s Speed of all vessels that will be deployed.

Problem $AUX(v, s, \kappa)$ has the following decision variables:

- N_j Binary variable, indicates whether port j is visited in the rotation.
- B_j Binary variable, indicates whether port j is a butterfly port in the rotation.
- I_j Continuous variable, indicates the sequence of port j in a rotation (for subtour elimination).
- C_j Binary variable, indicates if port j is the master port of the route.
- A_{ij} Binary variable, indicates whether edge (i, j) forms part of the new rotation.
- Q_{od} Continuous variable, indicates the number of FFEs with origin at port o and final destination at port d that will be carried per sailing of each vessel in the rotation.
- W_1, W_2 Binary variables, respectively, indicate whether the new rotation will have weekly or biweekly call frequency.
- μ Inverse of the number of trips to be completed over the entire planning horizon by each vessel on the new service.
- ω Estimated cost per sailing, per vessel, of the new service.

The auxiliary column generation model is then given by:

$AUX(v, s, \kappa)$

$$\text{maximize } Z_{AUX(v, s, \kappa)} = \omega \quad (11)$$

$$\begin{aligned} \text{subject to } \omega = & (\tilde{f}^v - f^v)\mu \\ & - \sum_{(i, j) \in \mathbf{E}} (eh^v p_j^v + eg^{vs} l_{ij}^v + d_j^v + a_{ij}^v) A_{ij} \\ & + \sum_{(o, d) \in \mathbf{G}} (q_{od} - u_o - u_d + \tilde{q}_{od}) Q_{od}, \end{aligned} \quad (12)$$

$$24T\mu = \sum_{(i, j) \in \mathbf{E}} \left(p_j^v + \frac{l_{ij}^v}{s} A_{ij} \right). \quad (13)$$

The objective function (11) maximizes the net revenue contribution from each sailing of the additional rotation. Constraints (12) define the contribution from each sailing by prorating the daily running cost of the vessel, the opportunity/layup cost of the vessel, sailing and hotel fuel expenses, port-calling fees, canal fees, and the revenues generated by the sailing. Constraints (13) establish the number of trips to be completed by each vessel over the decision horizon as a function of sailing time between ports and port stay duration.

The following constraints establish the basic logic for variables A_{ij} , N_i , and B_i :

$$\sum_{j \in \mathbf{P}} A_{jn} = \sum_{j \in \mathbf{P}} A_{nj} \quad n \in \mathbf{P}, \quad (14)$$

$$A_{ij} \leq (N_i + N_j)/2 \quad (i, j) \in \mathbf{E}, \quad (15)$$

$$\sum_{j \in \mathbf{P}} B_j \leq 1, \quad (16)$$

$$\sum_{i \in \mathbf{P}} A_{ij} \leq N_j + B_j \quad j \in \mathbf{P}, \quad (17)$$

$$\sum_{j \in \mathbf{P}} A_{ij} \leq N_i + B_i \quad i \in \mathbf{P}. \quad (18)$$

Constraints (14) balance the number of arcs entering and leaving any port in the rotation. Constraints (15) allow an arc (i, j) to be part of the rotation only if ports i and j are part of the rotation. The number of butterfly nodes in a rotation is limited to one by constraints (16). Constraints (17) and (18) limit the number of arcs that can enter or leave a port. If a port is a butterfly port, two arcs may enter and leave the port. Otherwise, a single entry and departure are permitted for each sailing.

To establish an approximate balance between the cargo that will flow through the new rotation and the capacity of the vessels deployed to the rotation, we have the following constraints:

$$Q_{od} \leq \mu \hat{k}_{od} / \kappa \quad (o, d) \in \mathbf{G}, \quad (19)$$

$$Q_{od} \leq \hat{k}_{od} N_o \quad (o, d) \in \mathbf{G}, \quad (20)$$

$$Q_{od} \leq \hat{k}_{od} N_d \quad (o, d) \in \mathbf{G}, \quad (21)$$

$$\sum_{d \in \mathbf{P}} Q_{od} \leq \phi_o^{\text{out}} c^v (N_o + \delta B_o) \quad o \in \mathbf{P}, \quad (22)$$

$$\sum_{o \in \mathbf{P}} Q_{od} \leq \phi_d^{\text{in}} c^v (N_d + \delta B_d) \quad d \in \mathbf{P}, \quad (23)$$

$$\sum_{(o, d) \in \mathbf{G}} l_{od}^v Q_{od} \leq \sum_{(i, j) \in \mathbf{E}} A_{ij} l_{ij}^v c^v. \quad (24)$$

Constraints (19) limit the amount of cargo that can be carried between any o – d pair by each vessel on each sailing, by prorating the total remaining demand among all vessels sailing in the rotation, and the number of trips to be performed by each vessel. Constraints (20) and (21) require ports o and d to be active in the rotation whenever any part of the residual demand for the corresponding o – d pair is carried by the candidate rotation. Our formulation does not represent the amount of cargo on board the vessels on every leg of the rotation. Rather, we use two approximations to balance the amount of cargo that is targeted for the proposed rotation against the capacity and number of vessels being deployed there. First, constraints (22) and (23) ensure that the amount of cargo loaded (respectively, discharged) at any port d is less than a certain fraction ϕ_d^{out} (respectively, ϕ_d^{in}), of the vessel capacity. The factors ϕ_d^{out} and ϕ_d^{in} capture the importance of each port as an importer or exporter of cargo relative to the overall residual demand for transport in the network. For the second approximation, we generate a lower bound (because we use the direct sailing distance for all o – d pairs) for the FFE miles of cargo being transported. For the second approximation, we estimate the FFE miles that are required by cargo being transported as well as the maximum FFE miles that can be provided by the new rotation. The left-hand side of constraints (24) represents a (rather weak) approximation on the FFE miles that will be consumed by the cargo being carried. The approximation is weak because we use the direct distance between origin and destination port l_{od}^v , whereas the cargo will more often travel indirectly, visiting several ports before reaching its destination. The right-hand side of constraints (24), however, provides an accurate measure of the FFE miles that will be provided by the new rotation.

The subtour elimination constraints are as follows:

$$\sum_{j \in \mathbf{P}} C_j = 1, \quad (25)$$

$$B_j \leq C_j \leq N_j \quad j \in \mathbf{P}, \quad (26)$$

$$N_j \leq I_j \leq |\mathbf{P}| N_j \quad j \in \mathbf{P}, \quad (27)$$

$$1 + I_i - |\mathbf{P}| C_j - |\mathbf{P}| (1 - A_{ij}) \leq I_j \quad (i, j) \in \mathbf{E}. \quad (28)$$

Constraints (25) identify exactly one port within the rotation as the base port. If the rotation has a butterfly

node, constraints (26) require that it must be at the rotation's base port. Constraints (27) force variables I_j away from zero if and only if the corresponding port is to be visited by the rotation. Constraints (28) force the value of sequence number I_j to increase along the path of the rotation, except when returning back at the rotation's base port.

To ensure that all rotations have weekly or biweekly frequency, we have the following constraints:

$$W_1 + W_2 = 1, \quad (29)$$

$$W_2 = 0 \quad c^v \geq 1,200 \text{ FFE}, \quad (30)$$

$$(W_1 - 1) + \frac{0.91 \cdot 7\kappa}{T} \leq \mu \leq \frac{7\kappa}{T} + (1 - W_1), \quad (31)$$

$$(W_2 - 1) + \frac{0.91 \cdot 14\kappa}{T} \leq \mu \leq \frac{14\kappa}{T} + (1 - W_2). \quad (32)$$

Constraints (30) reflect highly competitive conditions in markets where larger vessels are deployed. This constraint requires that rotations employing vessels with capacity at or above 1,200 FFE must have weekly calling frequency. It would be unwise to require rotation frequencies to be exactly 7 or 14 days. Given that the model uses discrete vessel speeds, a strict implementation of the (bi)weekly frequency requirement would likely discard many routes that are of commercial value, but that may not meet the target frequency by a few hours. Additionally, many carriers build some schedule slack into their routes to address fluctuations in the weather or delays at one of the ports in the rotation. With this in mind, we write constraints (31) so that rotations with nominal weekly frequency might in fact call as often as every 6.3 days. Similarly, constraints (32) permit that rotations with nominal biweekly frequency might call as often as every 12.7 days.

Finally, we have the integrality and nonnegativity constraints, as follows:

$$A_{ij} \in \{0, 1\} \quad (i, j) \in \mathbf{E}, \quad (33)$$

$$N_j, B_j, C_j \in \{0, 1\} \quad j \in \mathbf{P}, \quad (34)$$

$$W_1, W_2 \in \{0, 1\}, \quad (35)$$

$$Q_{od} \geq 0 \quad (o, d) \in \mathbf{G}, \quad (36)$$

$$I_j \geq 0 \quad j \in \mathbf{P}, \quad (37)$$

$$\mu, \omega \geq 0. \quad (38)$$

Our practical experience with problems of type $\text{AUX}(v, s, \kappa)$ is that these can be solved to optimality very quickly when the instance includes about 15 ports or less. Instances that entail more than 25 ports are significantly harder to solve, and are therefore not suitable for our overall strategy, where several thousand instances of $\text{AUX}(v, s, \kappa)$ may be launched. Our approach is to create clusters of ports that are tightly linked both geographically and by

trade volumes. We sort such clusters according to the total demand for transport between the cluster's ports, and select the top cluster. We then formulate problems $AUX(v, s, \kappa)$ using up to 20 ports from the selected cluster. We believe that this approach results in a good compromise between the speed (allowing us to generate many rotations in the heuristic column generation) and solution quality (finding the best rotations).

We also note that the subtour elimination constraints (28) are an important source of the difficulty in solving problem $AUX(v, s, \kappa)$. In an effort to build up the network very quickly when the algorithm starts, we suppress constraints (28) for 10 iterations. In many cases, we obtain solutions that do not contain subtours, allowing us to progress rapidly. As the network becomes more complex, however, it is necessary to reinstate constraints (28) to obtain valid solutions.

6.4. Complexity

In the following, we will prove that the LSNDP is strongly NP-hard by reduction from the travelling salesman problem (TSP) in the general case and the set-covering problem for the case of the model (2)–(10). Agarwal and Ergun (2008b) proved their model to be weakly NP-hard by reduction from the knapsack problem, so the proofs presented here are stronger.

In the general case, we may choose rotations arbitrarily. We show that LSNDP is strongly NP-hard by reduction from the TSP. The TSP may be defined as follows: Let $G = (N, A, C)$ be a graph where $N = \{1, \dots, n\}$ is the set of nodes, $A = \{(i, j) \mid i, j \in N, i \neq j\}$ is the set of edges, and $C = c_{ij}$ is a cost or distance matrix associated to the edges A . An optimal solution to the TSP is a minimal-cost Hamiltonian cycle covering the nodes in N .

THEOREM 6.1. *The LSNDP in the general case is NP-hard.*

PROOF. We reduce from TSP. Let the set of ports P correspond to the set of nodes N in the TSP, and let the travel cost between ports correspond to the cost matrix C between nodes in the TSP. Moreover,

set the demand between each pair of ports to 1 and limit the fleet list to one vessel of infinite capacity. The LSNDP will then choose the minimal-cost Hamiltonian cycle between the ports P to satisfy all demands, and is exactly an instance of the TSP. $\square$

If the rotations are given beforehand as in (2)–(10) it is easy to see that LSNDP is NP-hard by reduction from the set-covering problem. Given a number of nodes N and a family of sets $S_1, \dots, S_\ell$, $S_i \subseteq N$ with corresponding costs $\bar{c}_1, \dots, \bar{c}_\ell$, the set-covering problem asks to choose a subset of $S_1, \dots, S_\ell$ covering all nodes in N at the cheapest possible cost.

THEOREM 6.2. *The LSNDP based on fixed rotations is NP-hard.*

PROOF. To reduce the set-covering problem to LSNDP we let the set of ports be $\mathbf{P} = N \cup \{0\}$. For each subset S_i , we introduce a rotation $R_i = S_i \cup \{0\}$ that visits the ports in arbitrary order. The cost of a rotation is set to $\bar{c}_i = \bar{c}_i$. There is one vessel type, with ℓ vessels each having capacity $|N|$. All transshipment costs are set to 0, and there is a demand of 1 between each port $i \in N$ and port 0. The revenue of delivering one unit is $q_{od} = 0$, whereas the penalty of not delivering one unit is $\tilde{q}_{od} = \infty$. The LSNDP will now choose the cheapest set of rotations covering all ports, which is exactly the set-covering problem. $\square$

7. Computational Results

The model described in §6.1 has been solved heuristically for the LINER-LIB 12 data using the algorithm described in §6.2. The tests were performed on an Intel Xeon E5345, 2.66 GHz Quad core with 20 GB RAM. As LP solver we have used Gurobi 4.5. The running time of the algorithm varies because of the large difference in the size of the instances. The maximal running times have been set experimentally. As described in §5.6, there are three scenarios of each case representing high, medium, and low capacity related to the base case referred to as *low*, *base*, and *high*. In Table 7, column *t sec.* reports the running time per case and reports algorithmic performance

Table 7 Performance of the Algorithm and the Search Progress for the Median Run with Regard to the Objective Value

Instance	<i>t sec</i>	Columns evaluated			Unique columns			MCF eval		
		Low	Base	High	Low	Base	High	Low	Base	High
Baltic	300	12,912	12,586	14,046	1,521	1,719	1,866	13,680	13,255	14,809
WAF	900	13,120	15,343	12,248	3,858	4,637	3,951	13,481	15,343	12,602
Mediterranean	1,200	1,582	2,323	3,192	1,039	1,197	1,502	1,634	2,383	3,244
Pacific	3,600	4,987	4,203	4,726	2,552	2,361	2,660	5,060	4,268	4,769
AsiaEurope	14,400	4,882	4,291	4,889	2,322	2,215	2,806	4,972	4,357	4,942
WorldSmall	10,800	7,313	7,209	7,628	2,543	2,937	3,424	7,425	7,304	7,731
WorldLarge	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Notes. Instance denotes the instance, *t sec* is the running time in seconds. Columns evaluated is the number of rotations evaluated in the tabu search, Unique columns is the number of unique rotations, and finally, MCF eval is the number of times the multicommodity flow solution was calculated. Note that we do not solve WorldLarge because of its size.

Table 8 Performance of the Algorithm and the Search Progress for the Median Run with Regard to the Objective Value

Instance	Iter			Improving iter			Last improving iter		
	Low	Base	High	Low	Base	High	Low	Base	High
Baltic	548	358	459	12	12	9	151	30	12
WAF	267	337	271	19	30	26	115	195	221
Mediterranean	37	39	33	8	15	18	15	31	33
Pacific	57	54	33	31	26	30	57	30	33
AsiaEurope	102	80	87	64	60	58	102	80	85
WorldSmall	75	55	65	41	41	45	75	55	64
WorldLarge	NA	NA	NA	NA	NA	NA	NA	NA	NA

Notes. Instance denotes the instance, Iter is the number of iterations, where an iteration refers to $\mathbf{R}^u$, Improving iter is the number of iterations, where an improving solution was found, and finally, Last improving iter is the last iteration, where an improving solution was found.

for the median run of the *low*, *base*, and *high* cases. Ten replications of each instance have been made. The median is given as the fifth-worst value of the 10 runs. All figures and remaining tables display the best and

median solutions with regard to the objective function. Please note that a profit will be negative, because we are minimizing the objective value, but are in reality maximizing revenue.

Several parameters in the heuristic are model and algorithm dependent. The planning period is set to 180 days, and all demands are scaled accordingly. A weekly or biweekly frequency has been enforced in (31) and (32), and the first and second business rules of §4.2.5 are imposed. All bunker costs are fixed at 600 USD per metric ton. Port-draft incompatibilities are also considered in the model implementation.

Tables 7 and 8 report the algorithmic performance, Tables 9 and 10 report the objective values and cost components, and finally, some network key performance indicators are reported in Tables 11 and 12.

Figures 8(a) and (b) show the objective value for the median solution of the 10 randomized runs, as a function of the computational time, for the Baltic and AsiaEurope scenarios. For the remaining scenarios, refer to Appendix A.1.

Table 9 The Objective Value Separated by Cost and Income Components According to the Model in §6

Instance	Z	Q	c_v	c_b	c_p	c_c	c_m	c_t	L_v	F
Baltic										
Low										
Best	−6,044	96,750	7,740	9,522	8,808	0	53,780	0	0	10,850
Median	838	92,840	7,740	9,405	9,509	0	51,530	0	0	15,490
Base										
Best	−8,365	98,310	6,480	10,900	10,440	0	55,820	0	0	6,306
Median	−6,582	97,860	6,480	10,208	10,920	0	54,760	62.49	0	8,844
High										
Best	−15,678	103,500	5,760	11,460	10,520	0	59,140	0	0	942.2
Median	−11,509	101,200	5,760	9,894	11,670	0	57,240	159.4	0	4,954
WAF										
Low										
Best	−115,361	345,800	57,420	51,750	15,150	0	83,130	3,264	5,940	25,710
Median	−109,470	336,500	57,420	43,624	17,270	0	79,930	0	3,960	32,790
Base										
Best	−143,110	370,600	52,920	55,450	20,710	0	90,760	3,192	5,760	10,270
Median	−137,681	355,200	52,920	44,925	18,940	0	84,280	1,690	8,640	23,360
High										
Best	−159,944	382,000	48,960	60,520	16,910	5,827	94,830	3,859	10,800	1,956
Median	−152,635	367,800	48,960	48,756	19,160	5,827	88,650	1,376	11,880	14,330
Mediterranean										
Low										
Best	29,504	128,900	27,540	23,650	18,730	0	64,030	10,090	0	14,360
Median	42,636	121,700	27,540	27,546	17,320	0	62,000	9,694	0	20,270
Base										
Best	12,209	136,800	26,640	22,220	20,190	0	69,620	9,475	1,440	2,311
Median	24,934	134,200	26,640	23,591	20,610	0	68,430	16,630	1,980	5,187
High										
Best	6,606	137,100	26,100	22,070	19,330	0	69,920	9,861	5,400	1,807
Median	14,267	136,700	26,100	25,450	20,260	0	70,350	11,720	4,860	1,904

Notes. Instance denotes the name of the instance with a separate row for the best and median values over 10 replications. The best values correspond to the best solution obtained. Z is the objective value, note that we are minimizing expenses such that a negative value of Z is preferable; Q is the total revenue collected; c_v is the total vessel cost; c_b is the total fuel cost; c_p is the total port call cost; c_c are the canal fees; c_m is the total move cost at origin and destination nodes; c_t is the pure transshipment cost; L_v is the income from chartered out vessels; and finally F is the sum of goodwill penalties for rejected cargoes. All costs are in k\$. All numbers are presented with a four-digit precision. Z may deviate slightly from the sum of the remaining columns according to the objective because of this rounding.

Table 10 The Objective Value Separated by Cost and Income Components According to the Model in §6

Instance	Z	Q	c_v	c_b	c_p	c_c	c_m	c_t	L_v	F
Pacific										
Low										
Best	84,615	1,120,000	276,300	342,500	31,580	3,180	429,700	34,610	2,700	89,080
Median	132,020	1,044,000	276,300	300,400	32,460	2,989	390,100	31,450	0	142,700
Base										
Best	−54,087	1,197,000	247,700	348,900	31,610	6,101	451,600	34,870	13,500	35,440
Median	−12,774	1,160,000	247,700	351,800	35,210	2,967	444,400	29,110	17,460	53,710
High										
Best	−101,671	1,208,000	236,500	350,500	38,340	6,071	456,400	33,100	36,720	22,360
Median	−75,321	1,222,000	236,500	388,300	36,000	6,016	460,400	36,180	31,140	14,150
WorldSmall										
Low										
Best	−551,842	4,917,000	1,040,000	1,019,000	100,100	255,400	1,211,000	199,800	106,200	646,000
Median	−417,100	4,902,000	1,040,000	1,024,000	103,100	237,900	1,220,000	247,000	48,060	661,300
Base										
Best	−1,152,761	5,366,000	942,100	1,115,000	123,800	271,000	1,336,000	195,200	160,000	390,500
Median	−888,669	5,199,000	942,100	1,109,000	126,300	298,400	1,296,000	213,400	164,000	489,000
High										
Best	−1,291,718	5,550,000	911,700	1,202,000	129,300	287,800	1,377,000	263,600	223,200	310,400
Median	−1,168,067	5,266,000	911,700	1,077,000	109,400	267,100	1,319,000	244,000	261,500	431,800
AsiaEurope										
Low										
Best	−361,041	3,109,000	718,600	672,000	117,200	233,600	609,100	128,800	17,640	286,900
Median	−228,186	2,998,000	718,600	661,100	106,800	228,800	584,400	125,800	0	344,000
Base										
Best	−657,972	3,283,000	648,700	690,400	134,400	255,700	637,600	132,700	81,180	206,700
Median	−561,417	3,179,000	648,700	726,100	108,800	254,800	615,600	132,500	118,100	248,700
High										
Best	−766,385	3,390,000	625,700	834,900	130,000	276,700	656,000	127,900	176,600	149,400
Median	−670,099	3,181,000	625,700	708,900	117,700	253,900	622,400	131,400	184,900	235,300

Notes. Instance denotes the name of the instance with a separate row for the best and median values over 10 replications. The best values correspond to the best solution obtained. Z is the objective value, note that we are minimizing expenses such that a negative value of Z is preferable; Q is the total revenue collected; c_v is the total vessel cost; c_b is the total fuel cost; c_p is the total port call cost; c_c are the canal fees; c_m is the total move cost at origin and destination nodes; c_t is the pure transshipment cost; L_v is the income from chartered out vessels; and finally, F is the sum of goodwill penalties for rejected cargoes. All costs are in k\$. All numbers are presented with a four-digit precision. Z may deviate slightly from the sum of the remaining columns according to the objective due to this rounding.

Figure 8 shows that the objective value is converging for most cases—rapidly for the smaller instances—whereas for the larger instances there are still iterative improvements in spite of the increased run times. The ratio between unique columns (rotations) and total evaluated columns (rotations) is seen to increase with scenario size. This is because of the rotation generating MIP easily finding new unique columns in the larger search space of the larger instances, as opposed to the smaller instances. The Mediterranean instance is an exception, where the ratio is rather high. It is believed to be because of a much denser demand matrix compared to the Baltic and WAF case. The number of iterations refers to R^u of §6.2 and can be seen in Table 8. It can be seen that the number of iterations decreases with scenario size, even though computational time increases for the larger instances. It can be seen that the solution space grows significantly with instance size, because we find more improving solutions even in the last iterations.

Tables 9 and 10 state the objective value for the best and median cases. The total objective value of the instance is given by Z , and the remaining columns display separate costs for vessels, port calls, cargo handling, omission penalty, and incomes from chartering out vessels and transporting cargo. The difference in the objective value between the best and median cases shows a large variance in the resulting solution with regard to the objective value. It should be noted that the individual cost and income components of a large network are colossal, seen over a planning horizon of 180 days. The large variance may be caused by the heuristic getting trapped in a local minimum that it cannot escape. The solution space of liner-shipping network design is vast, and there are many individual components constituting a good solution. As a result, fixing a set of rotations to a particular vessel class may result in a profitable and well-utilized set of rotations, but a different composition of vessel class to the same rotations could result in a more profitable solution. However, the heuristic

Table 11 Key Performance Indicators for the Smaller Instances

Instance	Dep%	R	PCpW	BPU%	WPU%	BAU%	WAU%	$t/d\%$	Rej%
Baltic									
Low									
Best	100.0	5	2.81	100.0	100.0	93.9	47.9	0.0	8.6
Median	100.0	3	4.36	100.0	100.0	80.3	69.8	0.0	12.3
Base									
Best	100.0	4	4.06	100.0	100.0	75.9	43.2	0.0	5.0
Median	100.0	5	2.96	100.0	100.0	92.9	48.2	0.3	7.0
High									
Best	100.0	5	3.48	100.0	100.0	98.6	56.6	0.0	0.7
Median	100.0	4	4.12	100.0	100.0	83.6	61.5	2.7	3.9
WAF									
Low									
Best	89.4	7	5.19	100.0	95.1	90.4	50.4	7.0	11.7
Median	92.9	8	4.04	100.0	100.0	79.4	49.7	13.5	14.9
Base									
Best	88.9	11	3.69	100.0	95.9	82.9	44.6	16.5	4.7
Median	83.3	10	3.38	100.0	83.3	78.6	46.3	8.3	10.6
High									
Best	77.0	12	3.39	100.0	39.1	85.5	24.0	9.2	0.9
Median	74.7	10	3.64	100.0	39.1	93.1	24.0	1.6	6.5
Mediterranean									
Low									
Best	100.0	7	5.97	100.0	100.0	94.4	70.1	56.8	7.4
Median	100.0	7	5.57	100.0	100.0	95.3	76.9	50.4	10.4
Base									
Best	94.6	7	6.83	100.0	100.0	93.7	79.7	46.0	1.2
Median	91.9	8	5.68	100.0	90.2	90.4	62.3	84.0	2.7
High									
Best	78.4	11	4.75	100.0	76.1	96.3	60.5	48.8	0.9
Median	80.5	9	5.52	100.0	29.4	89.4	21.0	58.7	1.0

Notes. Instance denotes the name of the instance with a separate row for the best and median values over 10 replications. The best values correspond to the best solution obtained. The median value is the median value of all KPI over the 10 replications. Dep% is the percentage of the fleet deployed; |R| is the number of rotations in the final solution; PCpW is the average number of port calls per service per week; BPU%, WPU% is the best and worst peak utilization percentage, respectively; BAU%, WAU% is the best and worst average utilization percentage, respectively; $t/d\%$ is the percentage of transshipments performed of all delivered units; and Rej% is the percentage of rejected cargo.

will regard the set of rotations to be of high quality and will thus be less likely to remove them and try a different vessel class composition.

Tables 11 and 12 give some key performance indicators for the networks generated. The *utilization percentage* is the relation between cargo transported and the available capacity on a vessel for a single voyage between two ports. The *peak utilization* of a rotation relates to the voyage between two ports with the highest utilization percentage. *Best peak utilization* (BPU) denotes the peak utilization of the rotation with the highest peak utilization, and *worst peak utilization* (WPU) is the rotation with the lowest peak utilization. The *average utilization* relates to the average utilization percentage of all voyages on a rotation. *Best average utilization* (BAU) describes the average utilization of the rotation with the highest average utilization, and *worst average utilization* (WAU) is the average utilization of the rotation with the lowest average utilization.

Fleet utilization and rejected demand. The percentage of the fleet deployed *dep%* seen in relation to the

percentage of rejected demand *rej%* reveals lack of capacity to carry all demand for the low-capacity instances as expected. Likewise, there is excess capacity in most high-capacity instances with a corresponding low rejection rate. AsiaEurope and WorldSmall high instances are an exception to this pattern, which may be due to the profitability of cargo, but because the large instances are continuously finding improving rotations, solutions with a higher deployment and a decrease in rejected demand may exist.

Number of rotations. The number of rotations are seen to increase with instance size. WorldSmall has a larger number of rotations than AsiaEurope correlated to the number of average port calls on a service, PCpW being lower in WorldSmall than in AsiaEurope.

Utilization. All solutions have a BPU of 100%, meaning that at least one service is fully utilized on one voyage between two ports. Low WPUs are also seen and can be acceptable for feeder services for outlying profitable cargo. Looking at the average utilization, BAU%, WAU%, the networks are overall well utilized because some variation must be expected given the

Table 12 Key Performance Indicators for Larger Instances

Instance	Dep%	R	PCpW	BPU%	WPU%	BAU%	WAU%	$t/d\%$	Rej%
Pacific									
Low									
Best	99.0	22	5.41	100.0	87.4	100.0	50.0	42.3	7.8
Median	100.0	21	5.09	100.0	32.8	100.0	18.8	41.7	12.6
Base									
Best	95.0	21	5.73	100.0	36.6	100.0	21.0	46.0	3.1
Median	93.3	21	5.54	100.0	58.8	100.0	38.5	44.8	4.7
High									
Best	85.1	20	6.53	100.0	100.0	100.0	50.0	47.5	2.0
Median	87.0	24	5.02	100.0	7.7	100.0	5.3	39.0	1.2
WorldSmall									
Low									
Best	90.1	35	5.60	100.0	100.0	100.0	50.9	64.7	19.6
Median	95.5	42	4.04	100.0	48.7	100.0	42.7	74.7	20.0
Base									
Best	83.0	41	4.87	100.0	37.0	100.0	28.3	64.7	11.8
Median	82.8	40	4.67	100.0	22.1	100.0	21.1	65.3	14.8
High									
Best	76.5	41	4.68	100.0	24.0	100.0	23.1	73.7	9.4
Median	71.9	37	5.06	100.0	79.6	100.0	52.7	78.3	13.1
AsiaEurope									
Low									
Best	97.5	30	6.39	100.0	34.0	98.5	22.4	80.2	14.5
Median	100.0	32	5.50	100.0	32.5	100.0	25.7	75.3	17.4
Base									
Best	87.0	35	6.59	100.0	34.9	96.0	32.2	91.1	10.4
Median	81.1	38	5.48	100.0	22.9	100.0	16.0	81.6	12.6
High									
Best	71.0	37	6.29	100.0	11.1	100.0	7.5	81.5	7.6
Median	70.6	34	6.28	100.0	33.4	100.0	23.7	82.9	11.9

Notes. Instance denotes the name of the instance with a separate row for the best and median values over 10 replications. The best values correspond to the best solution obtained. The median value is the median value of all KPI over the 10 replications. Dep% is the percentage of the fleet deployed; |R| is the number of rotations in the final solution; PCpW is the average number of port calls per service per week; BPU%, WPU% is the best and worst peak utilization percentage, respectively; BAU%, WAU% is the best and worst average utilization percentage, respectively; $t/d\%$ is the percentage of transshipments performed of all delivered units; and Rej% is the percentage of rejected cargo.

asymmetry of world trade and low utilization in some parts of the feeder network.

Transshipment percentage. The percentage of the number of transshipments over the demand units transported $t/d\%$ reveals that Baltic and WAF instances have very few transshipments and, as expected, the Mediterranean case has a larger percentage of transshipments. For Pacific, WorldSmall, and AsiaEurope most demands are subject to one or more transshipments.

7.1. Discussion

Overall, the solutions obtained are promising. The solutions utilize the available capacity well, and the constructed networks transport the majority of the profitable cargo, as can be seen in Tables 11 and 12.

Review by Network Planners. Network planners at Maersk Line have evaluated that the given solutions are feasible liner-shipping networks for the scenarios, but with room for improvement by inspection of

experienced eyes. Some of these are dealt with by the incorporated business rules, but others such as cabotage rules, commercially-driven transit time restrictions, etc., are not dealt with. Realistic traits of the networks can be seen in the construction of shorter feeder services and longer intercontinental services. Hubs are used realistically by connecting several feeders and intercontinental services to the same hubs.

Baltic and WAF. These small instances show very fast convergence in Figures 8 and A.1. The generated networks consist of services with few port calls (Table 11) (except lowly capacitated *low* WAF instances), as expected of feeder networks with few or no transshipments.

Mediterranean has a dense demand matrix which facilitates some hub structure and transshipments. No runs generated a profitable solution, but the high-capacity cases are close, the best with a loss of 6,606 k\$. A study of the revenues of the demands reveal a rather low revenue compared to, e.g., the Baltic case, and

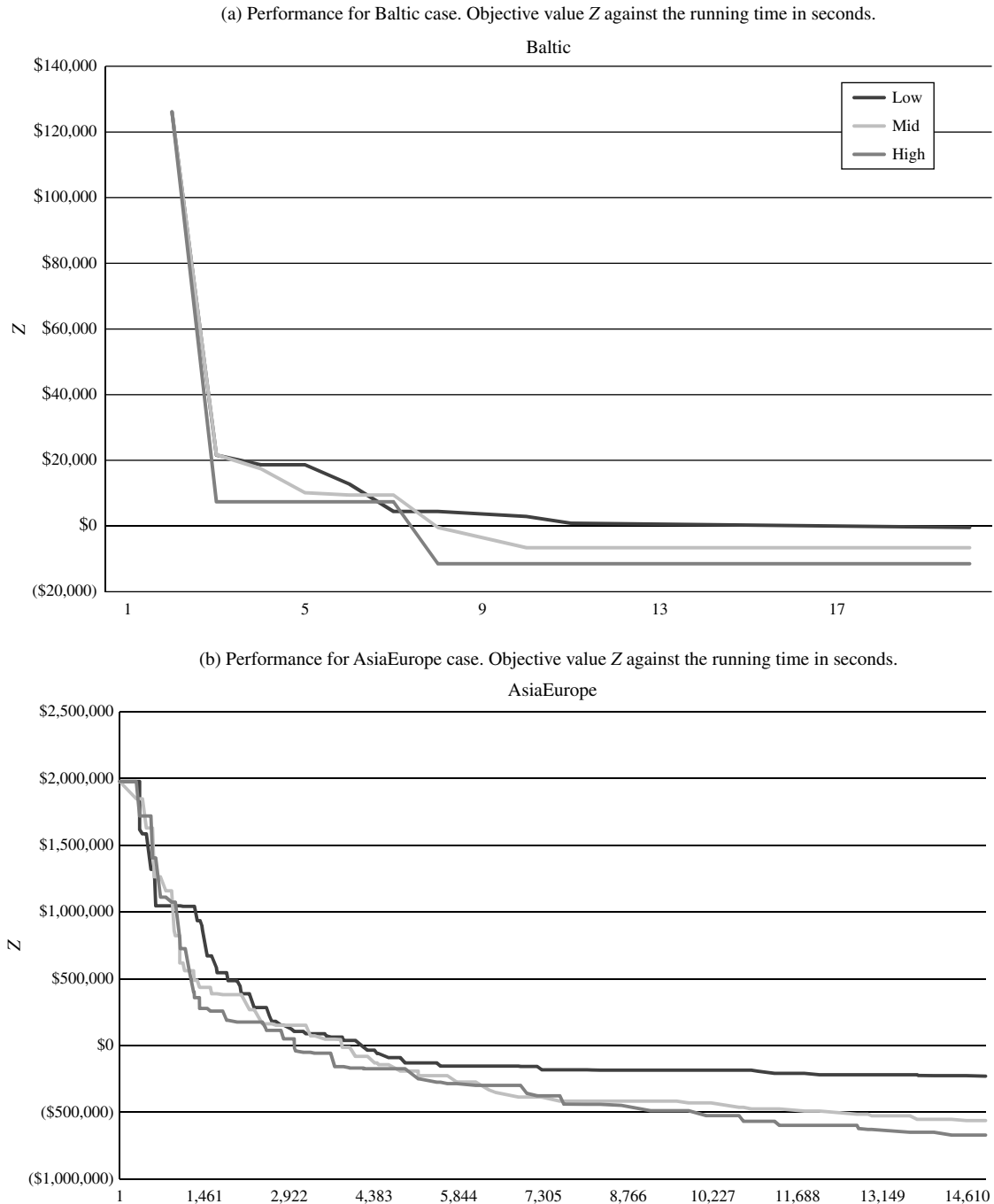


Figure 8 The Objective Value for the Median Run Solution of the Low, Base, and High Instances of the Baltic and AsiaEurope Case, as a Function of the Running Time

also the actual demand is spread on a large number of demand pairs, meaning that the cargo compositions are more complex. The analysis indicates that the Mediterranean case is not very attractive from an economic standpoint.

Pacific and WorldSmall. For these large instances, complex networks are created with 20–40 services calling on five to six ports on average. For Pacific, most cargo is transported, except for the lowly

capacitated instances; for the WorldSmall, a little less is moved, but as vessels are chartered out it must be due to unprofitable demand, but still it is the most complex scenario, and improvements are possible. The Pacific case has relatively few transshipments, because of the direct connections over the Pacific and the high North American transshipment costs. WorldSmall has realistically massive use of transshipments.

AsiaEurope. The AsiaEurope case is a large challenge due to the size of the instance, but also because the revenue is tight. From the utilization indicators (BAU/WAU) in Tables 11 and 12, it can be concluded that traffic flow is very intense in one direction, displaying the trade imbalance, which is very decisive for network design today because of the excess capacity on the backhaul. This case is well suited for future work optimizing on empty repositioning. The amount of rejected cargo of the larger instances clearly indicates that some demand is in reality loss giving, a trait any real-life case will have.

Cost Structure. Looking at the median solution for the largest base capacity scenario, WorldSmall, the network cost is distributed as 24% for vessel costs c_v , 28% for fuel costs c_b , 10% for port call and canal costs $c_p + c_c$, and 38% for move cost $c_m + c_t$. Noting that this is a simplified case with demand aggregated to larger ports, reducing the needed number of transshipments and port calls, this cost distribution can resemble the cost distribution of real liner-shipping networks, although this of course fluctuates with cost and market changes.

Transshipments and Hubs. The cases vary greatly in the hub-and-spoke structure, and hence the desire to transship cargo. The Baltic case has very few transshipments, as will be the case for a feeder network. In the AsiaEurope case all cargo transship more than once on average. Seen from a commercial point of view, a transshipment carries a risk in terms of disruptions and an administrative overhead in transshipped cargo. These perceived “costs” of transshipping is not considered in the mathematical model, and the found solution may have excess transshipments compared to real life. Adding customer satisfaction costs to the transshipment costs in the model can to some extent account for this problem.

Service Types. The solutions clearly display the capacity levels of a feeder network and an intercontinental network, where larger vessels are servicing intercontinental routes. Network planners at Maersk Line confirm that the structure of the networks in terms of capacity deployment and the use of transshipment locations is fairly coherent with the operations of Maersk Line. Also, the rotations vary greatly in design and display all of the design structures seen in Figures 1 and 2, showing the algorithms adaptability for different situations.

Fleet Size. The different fleet-size instances low, base, and high capacity result in different types of networks. The low-capacity instances obviously have fewer services, which on average are better utilized, because they will fill the vessels with the most profitable cargo and reject more cargo. The high-capacity instances will have to create additional services to make a profit of the remaining cargo. Thus, the

resulting networks of the related fleet-size scenarios will result in quite different networks and add to the complexity of LINER-LIB 12.

8. Conclusion and Future Work

In this paper we have given an introduction to liner-shipping network design for operations researchers. The problem domain has been described, and a discussion of mathematical modelling of the domain and the constraint set has been provided. A set of data instances was introduced as a benchmark suite, resembling real-world problems. The problem has been proven to be strongly NP-hard, and compared with optimal methods for solving related liner-shipping network design models, it is apparent that this problem is among the most difficult network design problems. The model of Alvarez (2009) has been extended to handle the structure of a complex network, and a heuristic column generation algorithm has been presented using a new method for route generation. Computational results using LINER-LIB 12 have been presented, and the solutions will be made available on a webpage to encourage a competition in developing the best algorithms. The algorithmic performance indicators suggest that we have created a benchmark suite that may be used for the development of both exact and heuristic approaches because the smaller cases should be suitable for benchmarking exact algorithms and the large instances pose a substantial challenge for state-of-the-art heuristics. Overall, LINER-LIB 12 provides a thorough test of a network design algorithm. The cases challenge the fleet deployment, port call sequence generation, the transshipment structure of the network, and the selection of demand to transport in the network. At the same time, the cases created closely resemble the real-life operations of a global liner-shipping company and the market for liner shipping in terms of trade imbalance, geographical transshipment points, and fluctuations in demand compared to the capacity available. We also believe that LINER-LIB 12 will prove valuable to future model development of the liner-shipping network design problem. There is a large potential for future work on modelling because several commercial requirement, such as low transit times and slow steaming, have not been incorporated in the current network design models. The results have been evaluated by network designers of Maersk Line, who affirm that the traits of individual rotations resemble real-world rotations, and that the overall network fulfills most of the relevant properties. It is also apparent by inspection that the heuristic solutions may be improved upon, which encourages the development of better solution methods. We hope that this work will serve as a foundation for unifying and encouraging further work in liner-shipping network design,

which is a challenging real-world problem having a huge impact on the cost of the world’s supply chains. Moreover, the scientific world needs complex challenges to push algorithm development forward.

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Appendix A. Figures

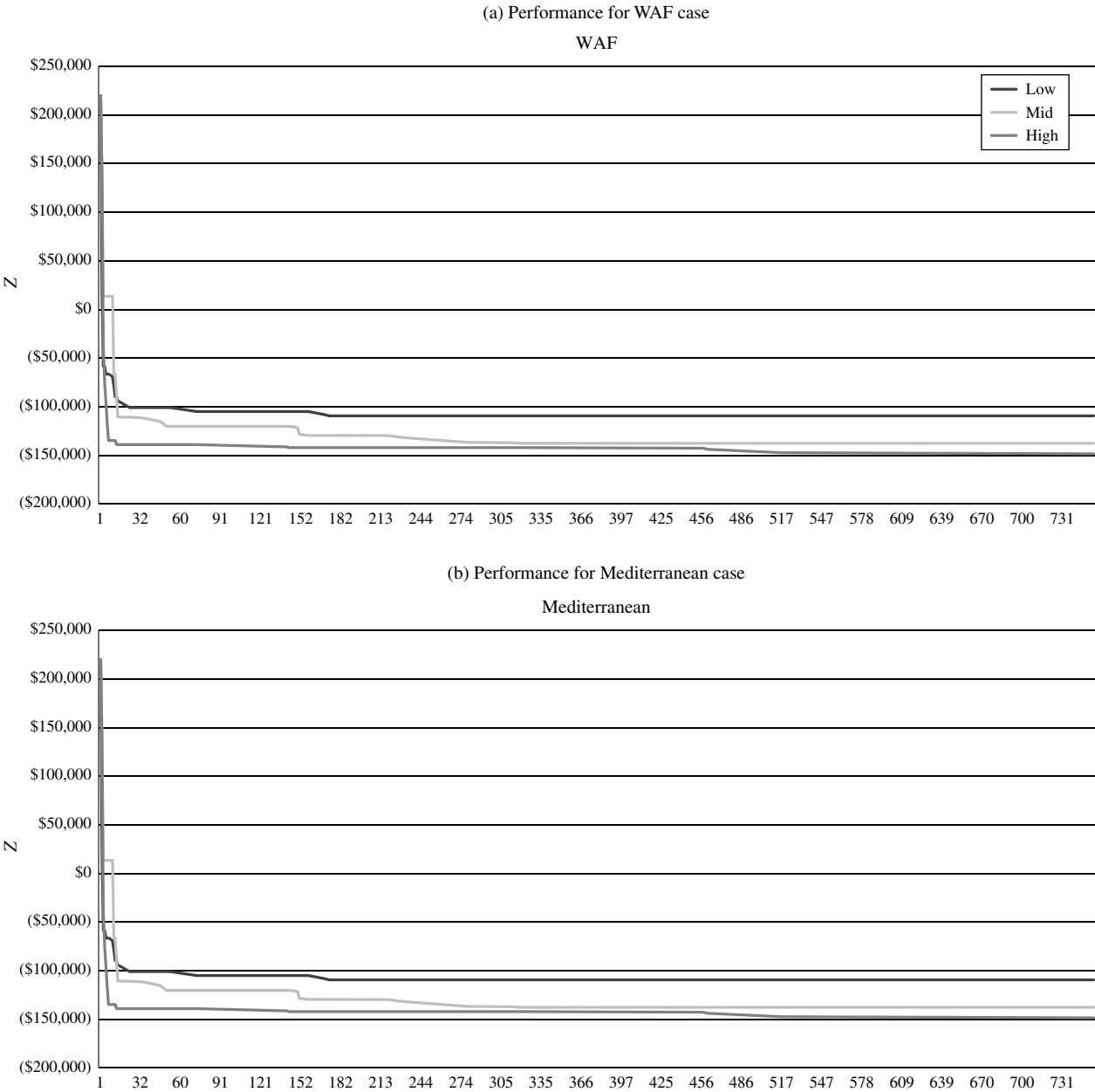


Figure A.1 The Objective Value for the Median Run Solution of the Low, Base, and High Instances, as a Function of the Running Time in Seconds (WAF and Mediterranean Case)

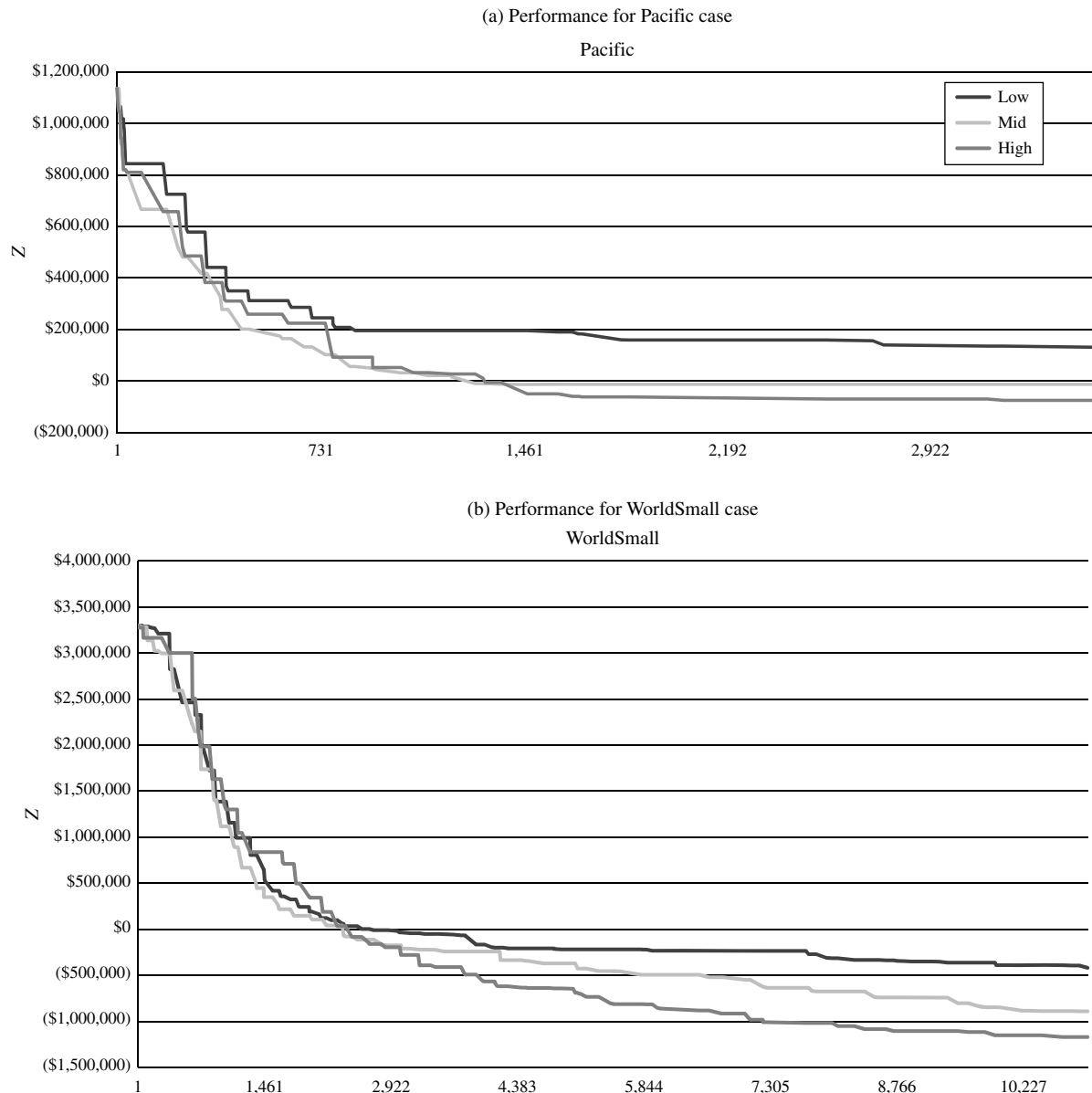


Figure A.2 The Objective Value for the Median Run Solution of the Low, Base, and High Instances, as a Function of the Running Time in Seconds (Pacific and WorldSmall Case)

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